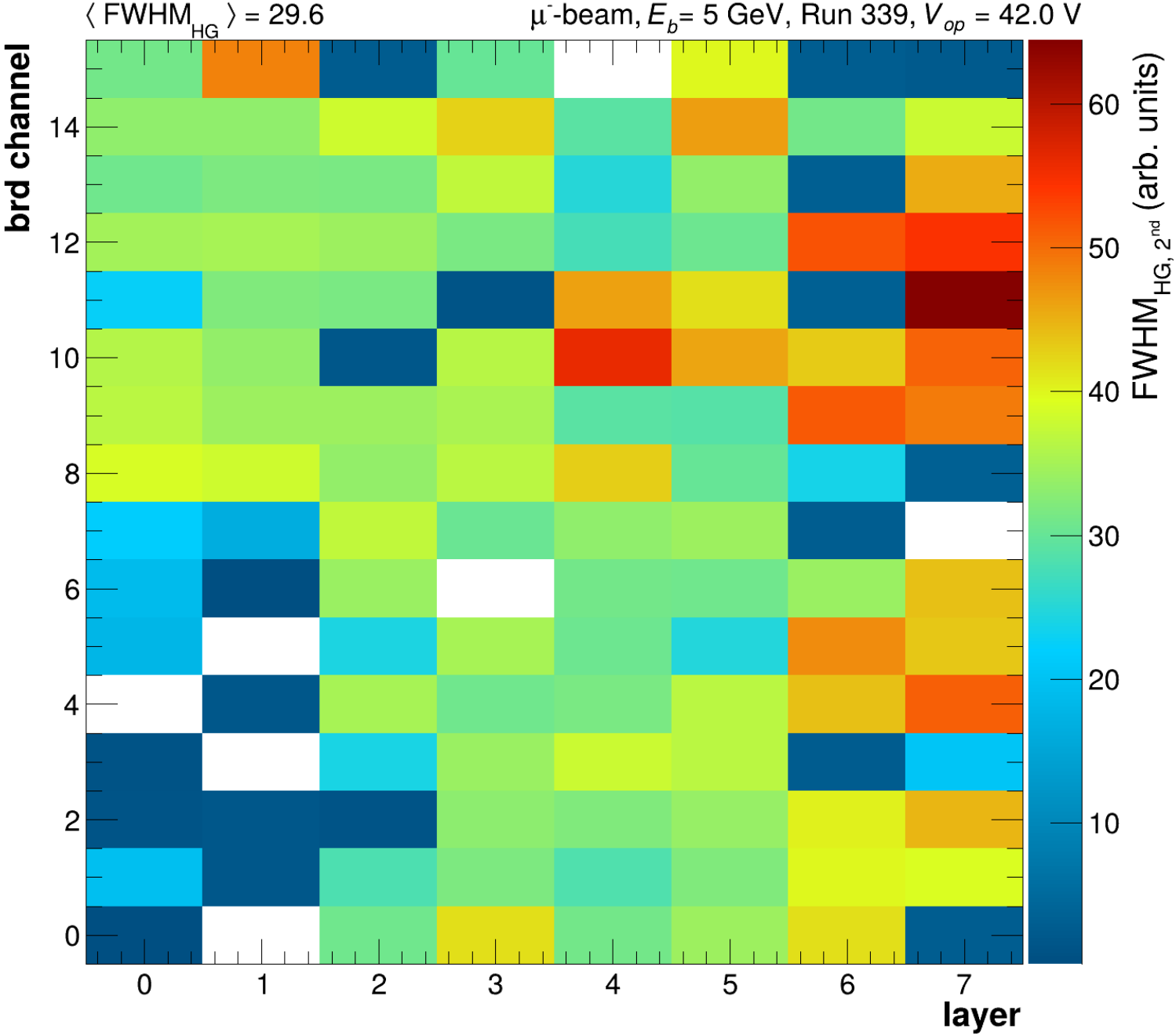
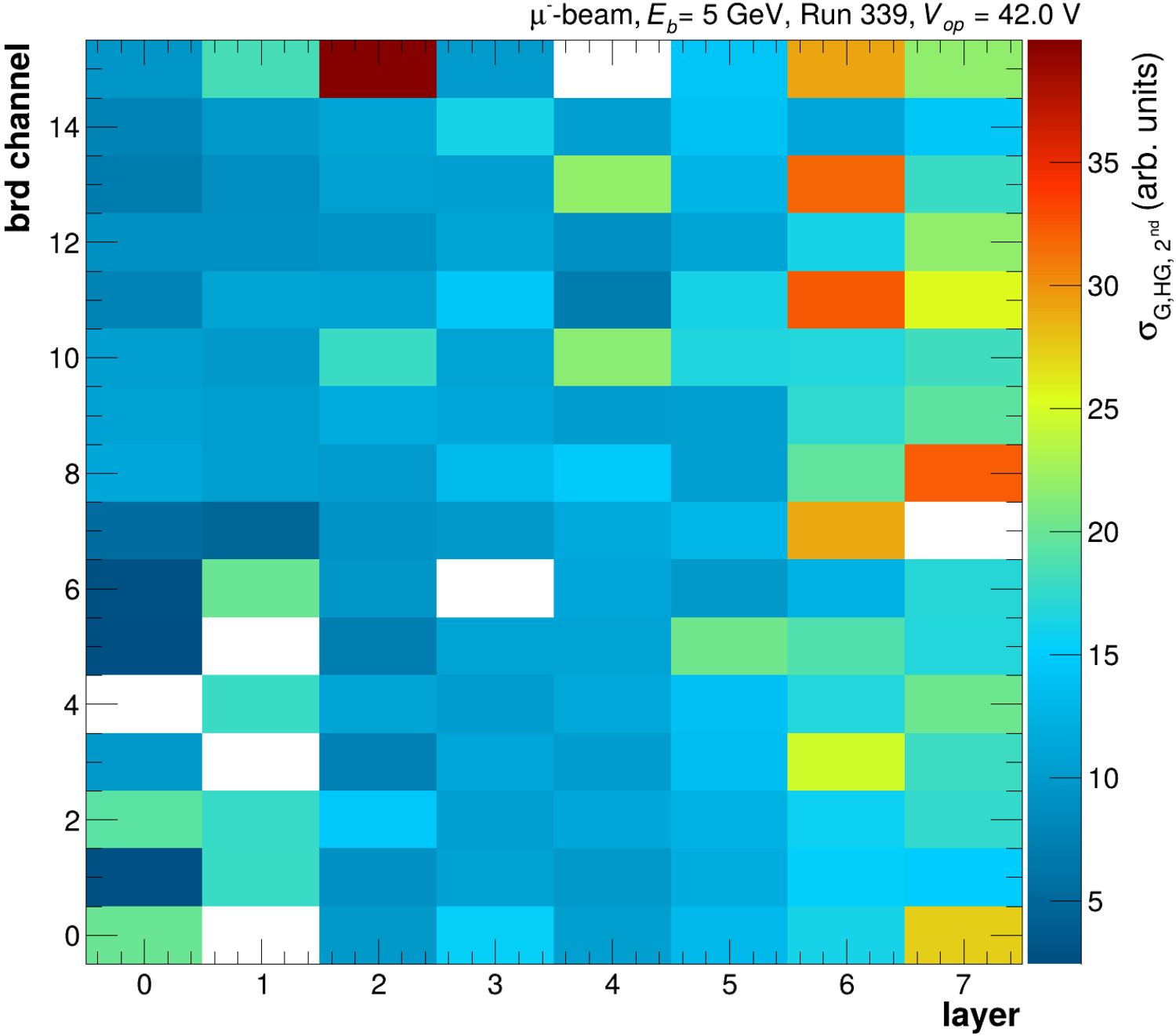
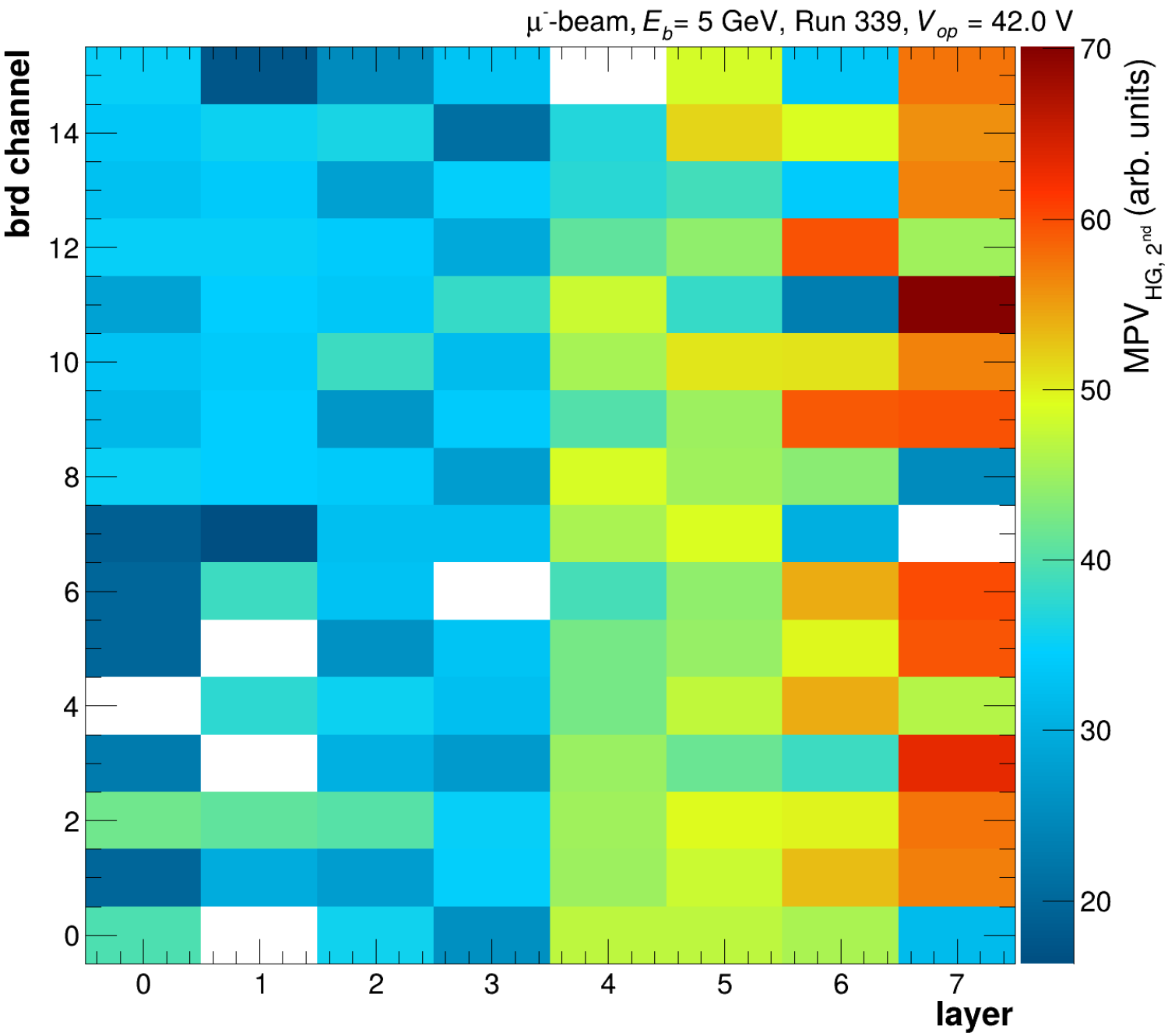
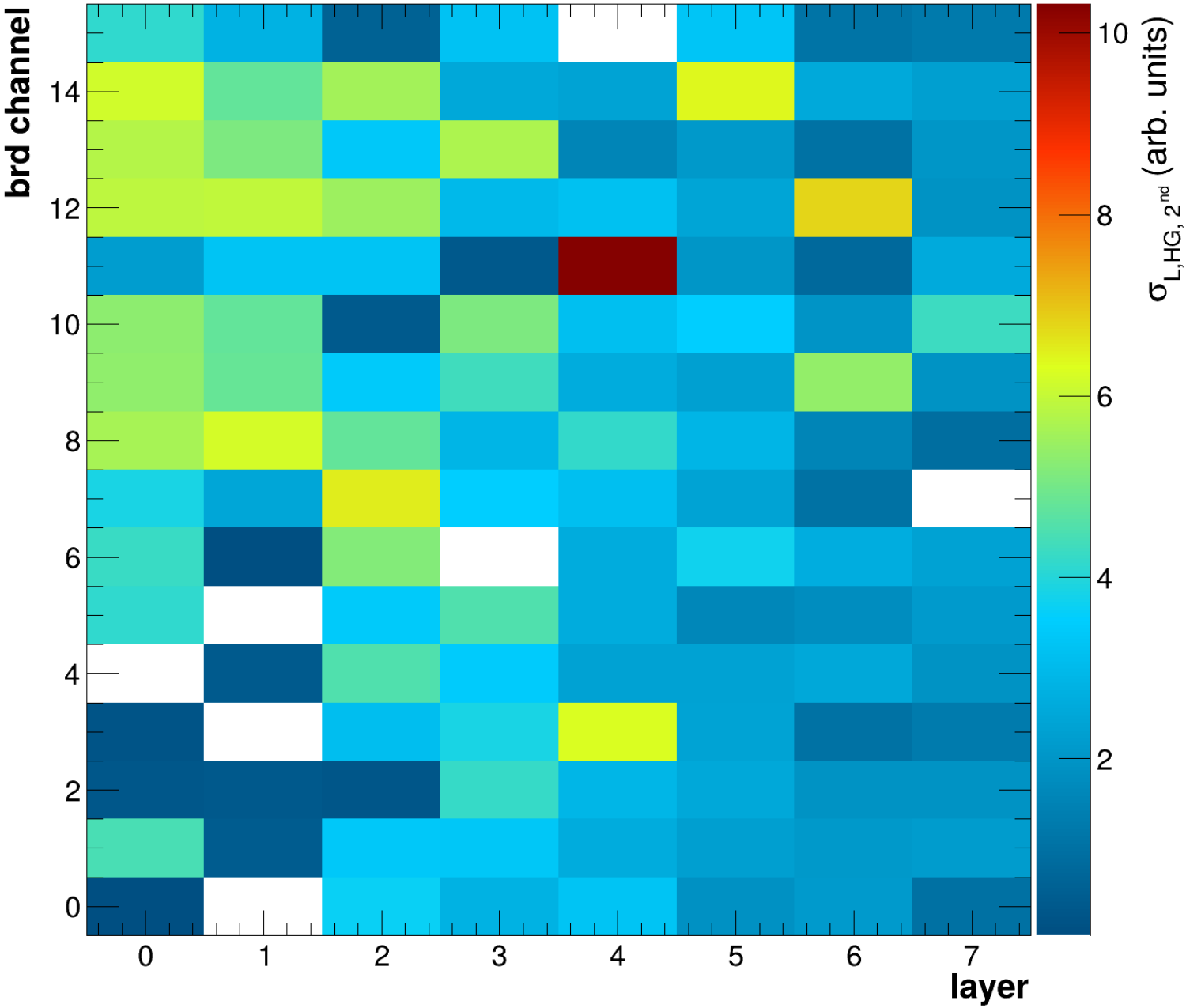
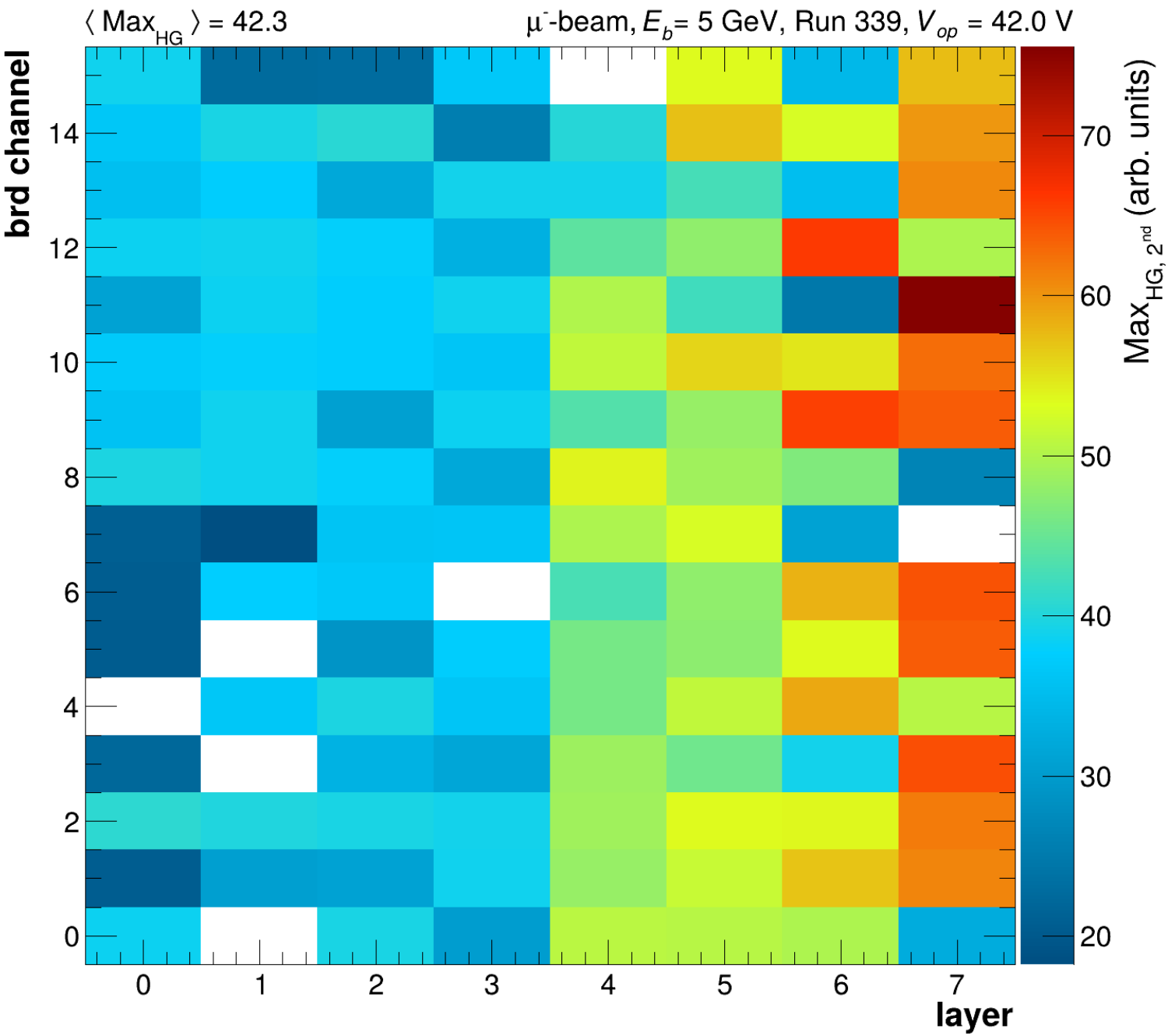
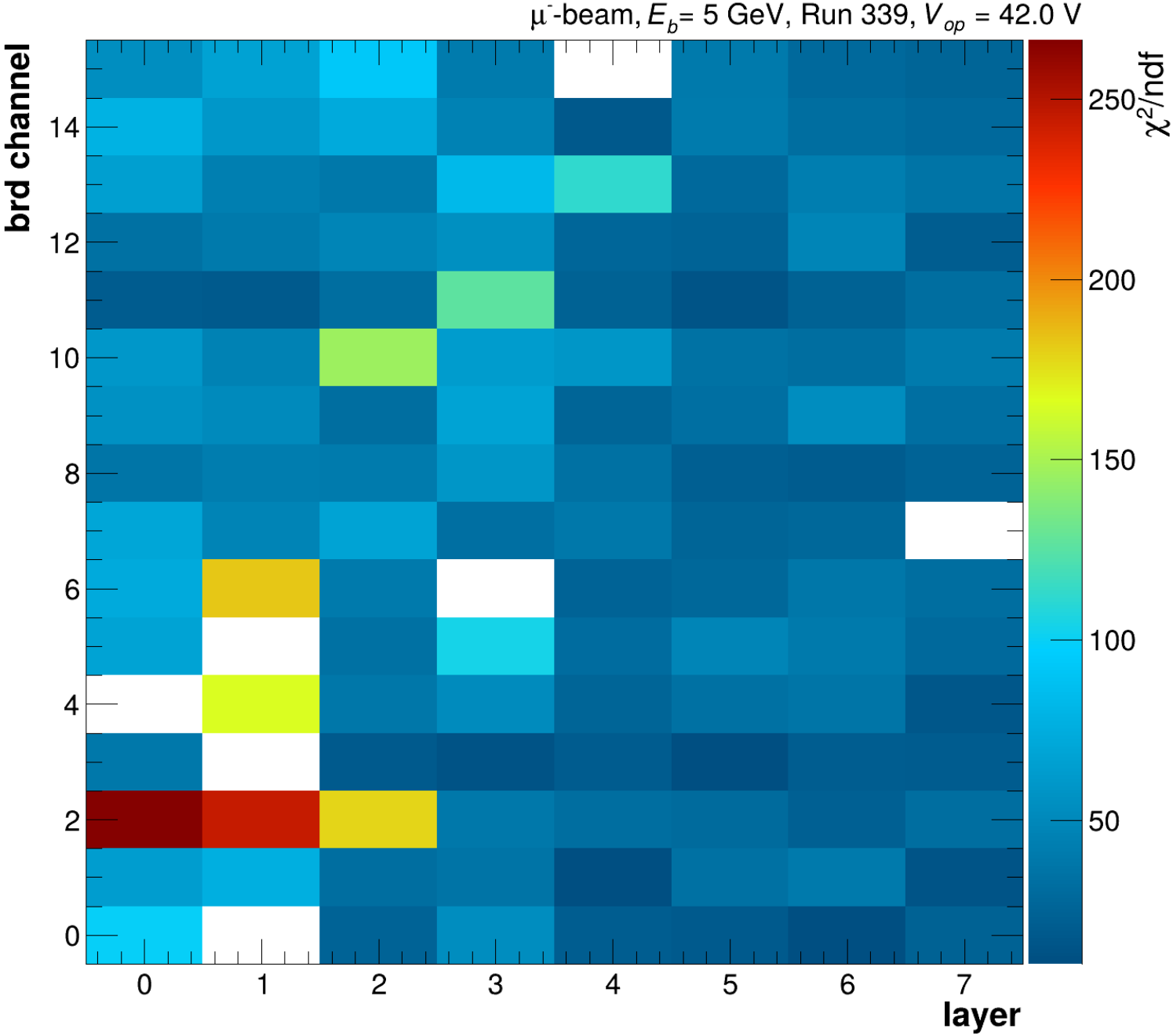

⟨ FWHM_HG ⟩ = 29.6
μ⁻-beam, E_b= 5 GeV, Run 339, V_op = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
FWHM_HG, 2nd (arb. units)
10
20
30
40
50
60

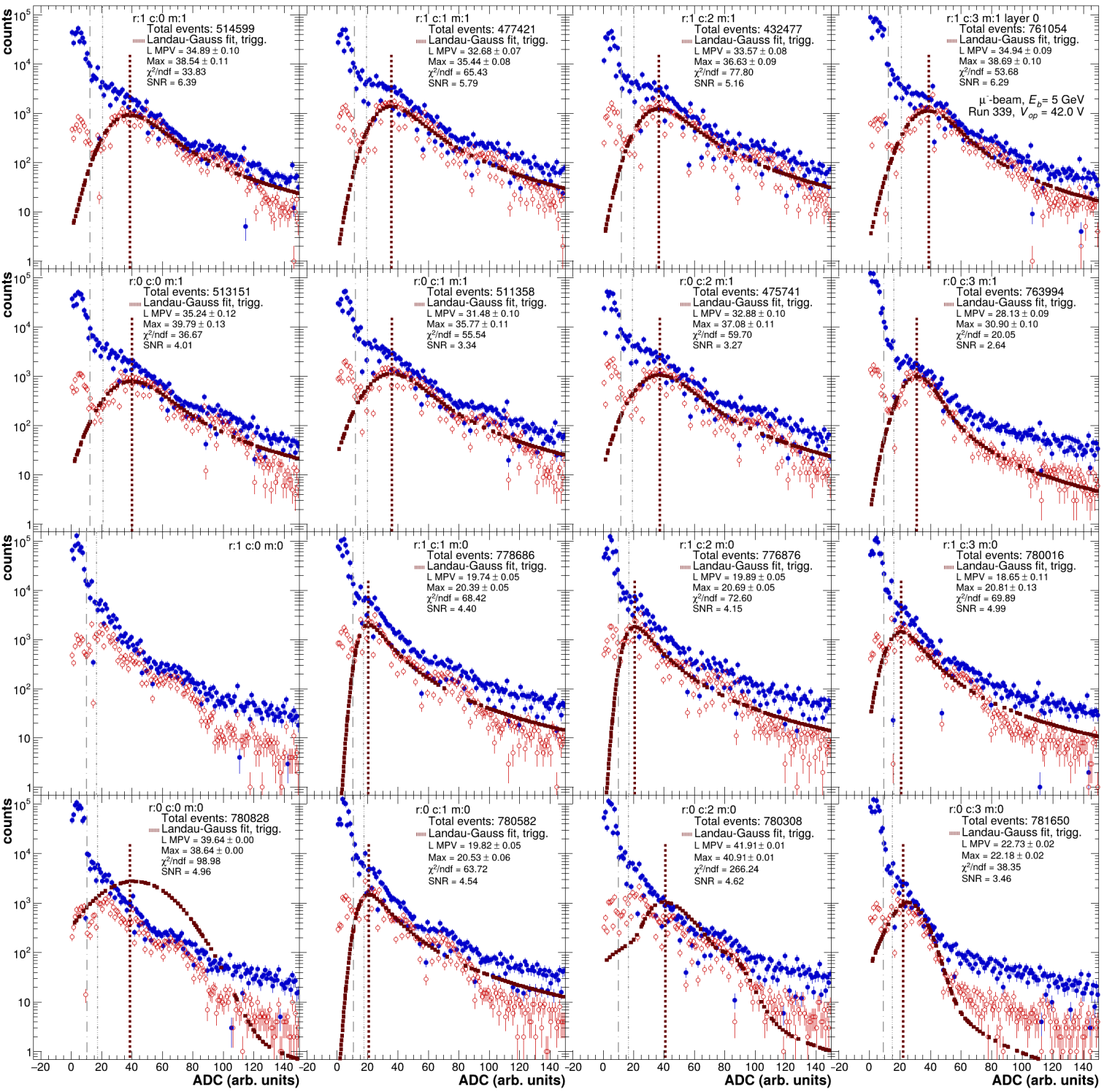

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35

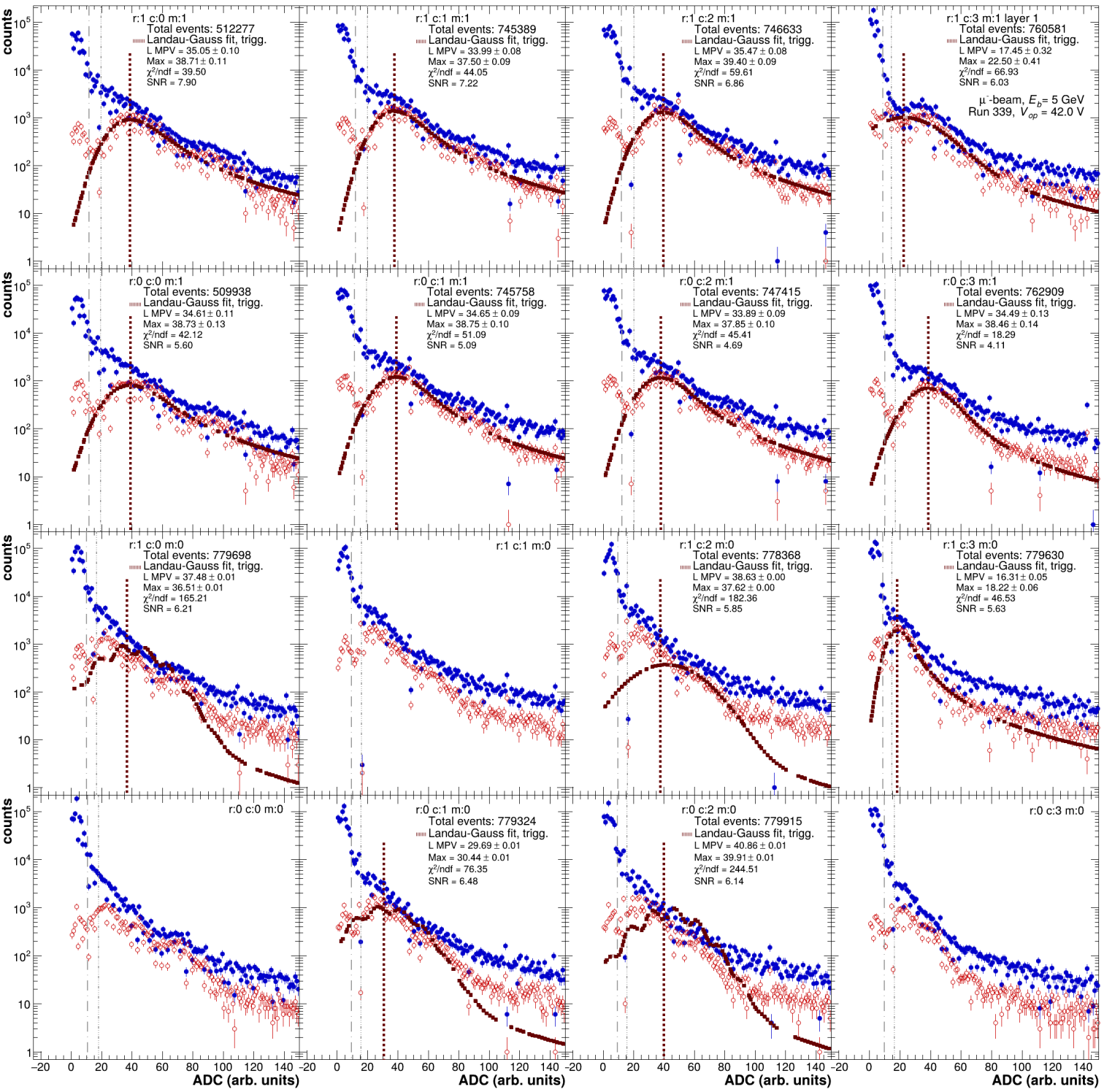

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
20
30
40
50
60
70

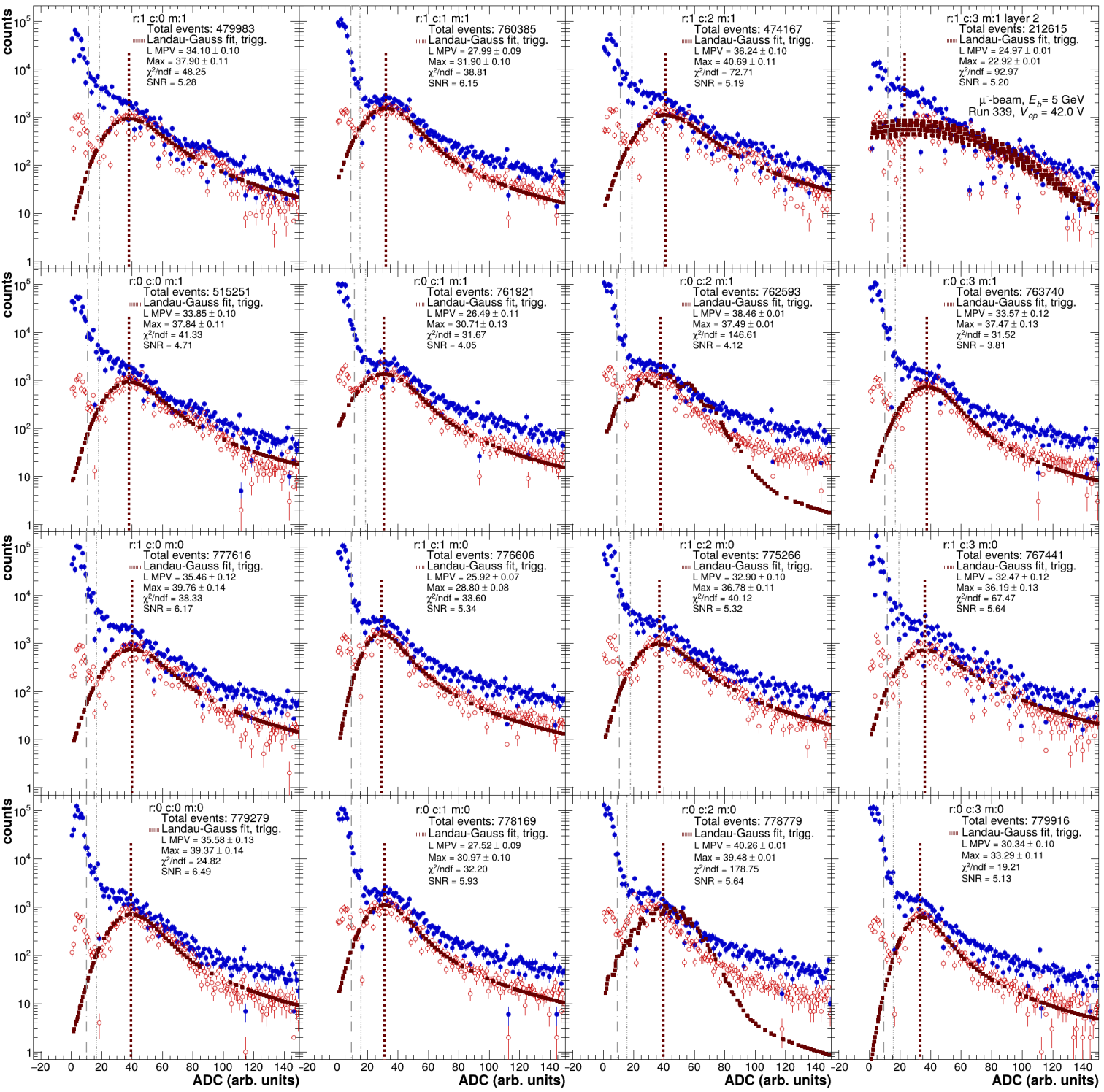

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

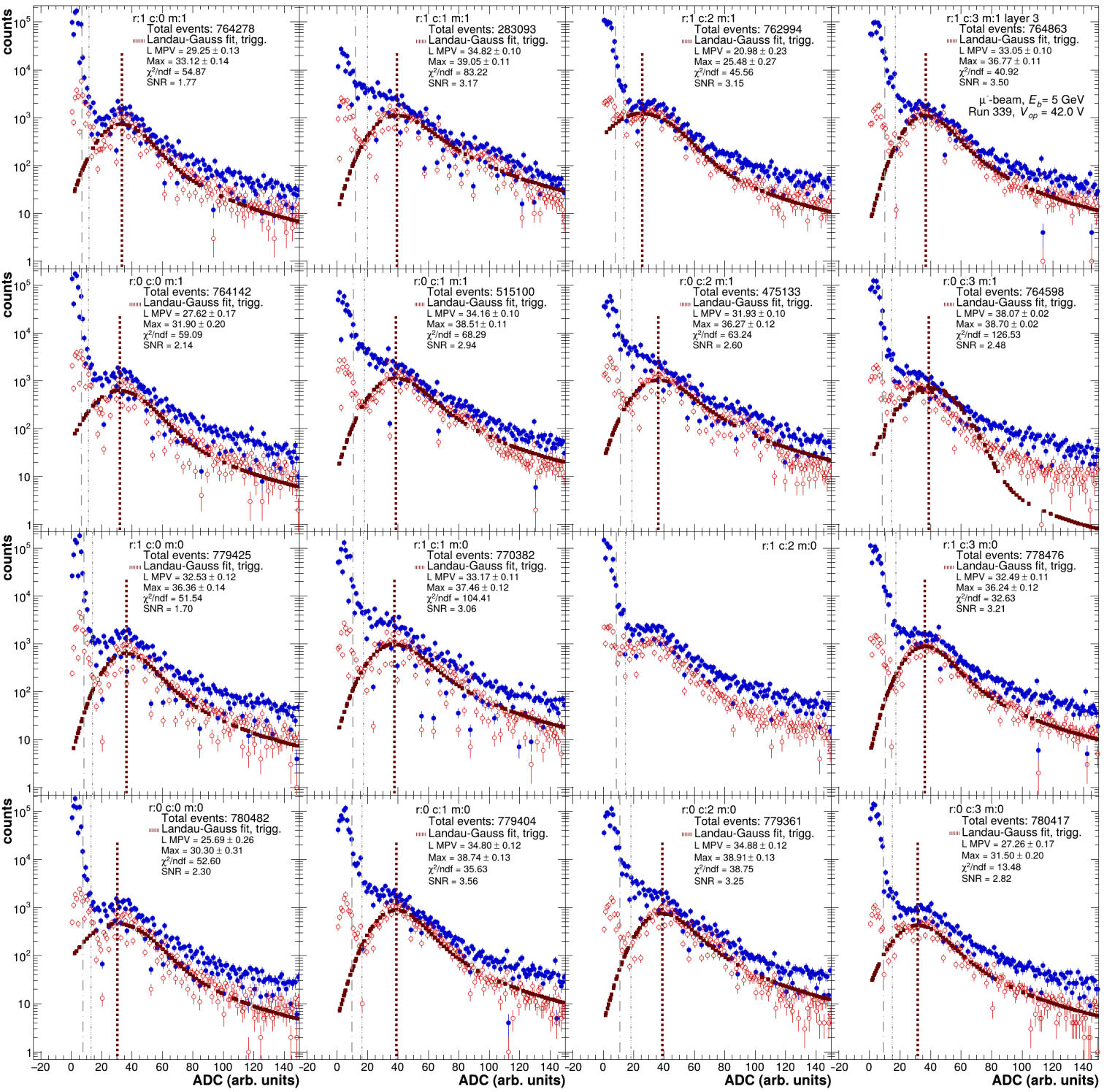

⟨ Max_HG ⟩ = 42.3
μ⁻-beam, E_b = 5 GeV, Run 339, V_op = 42.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
30
40
50
60
70

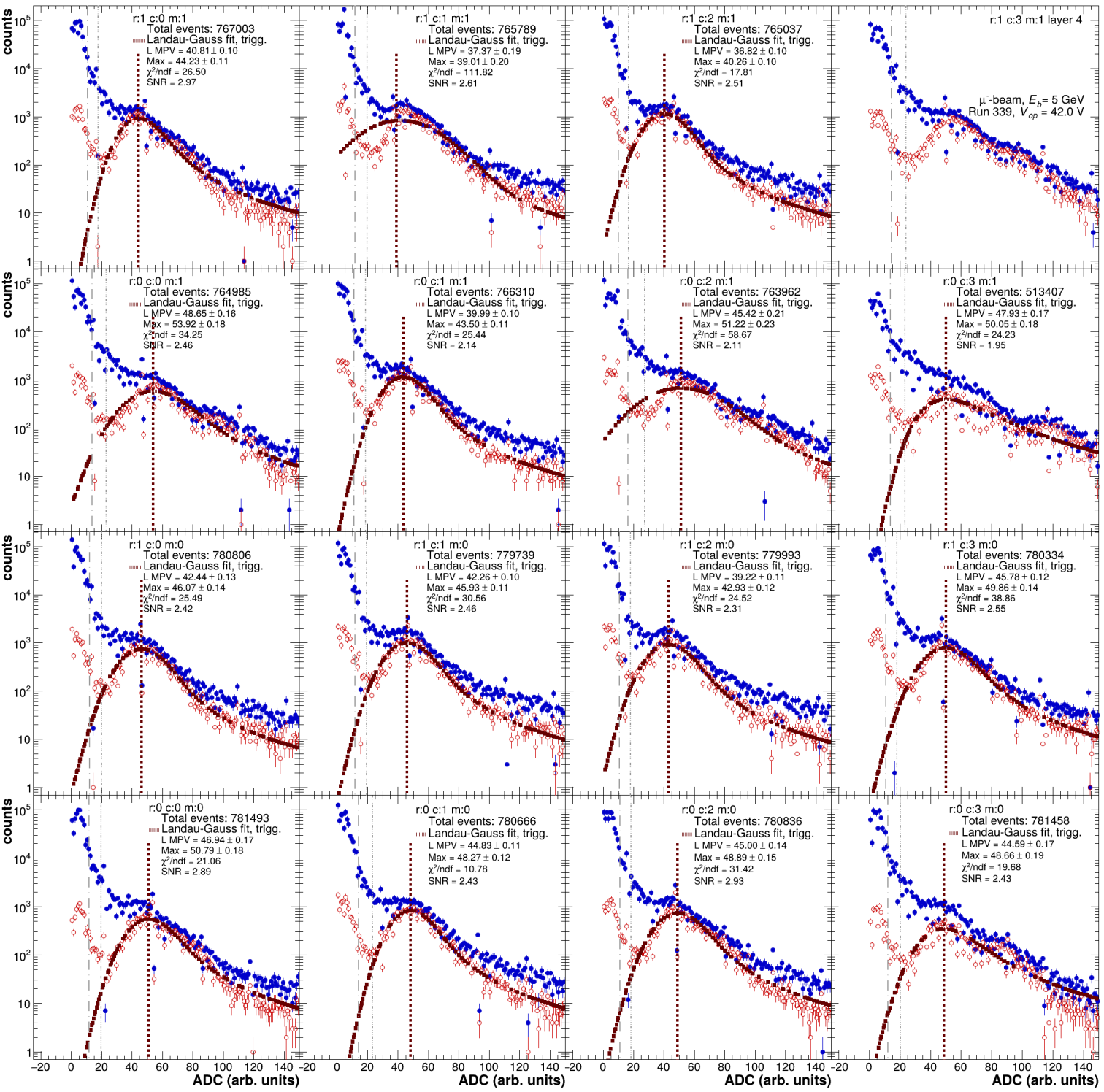

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
50
100
150
200
250

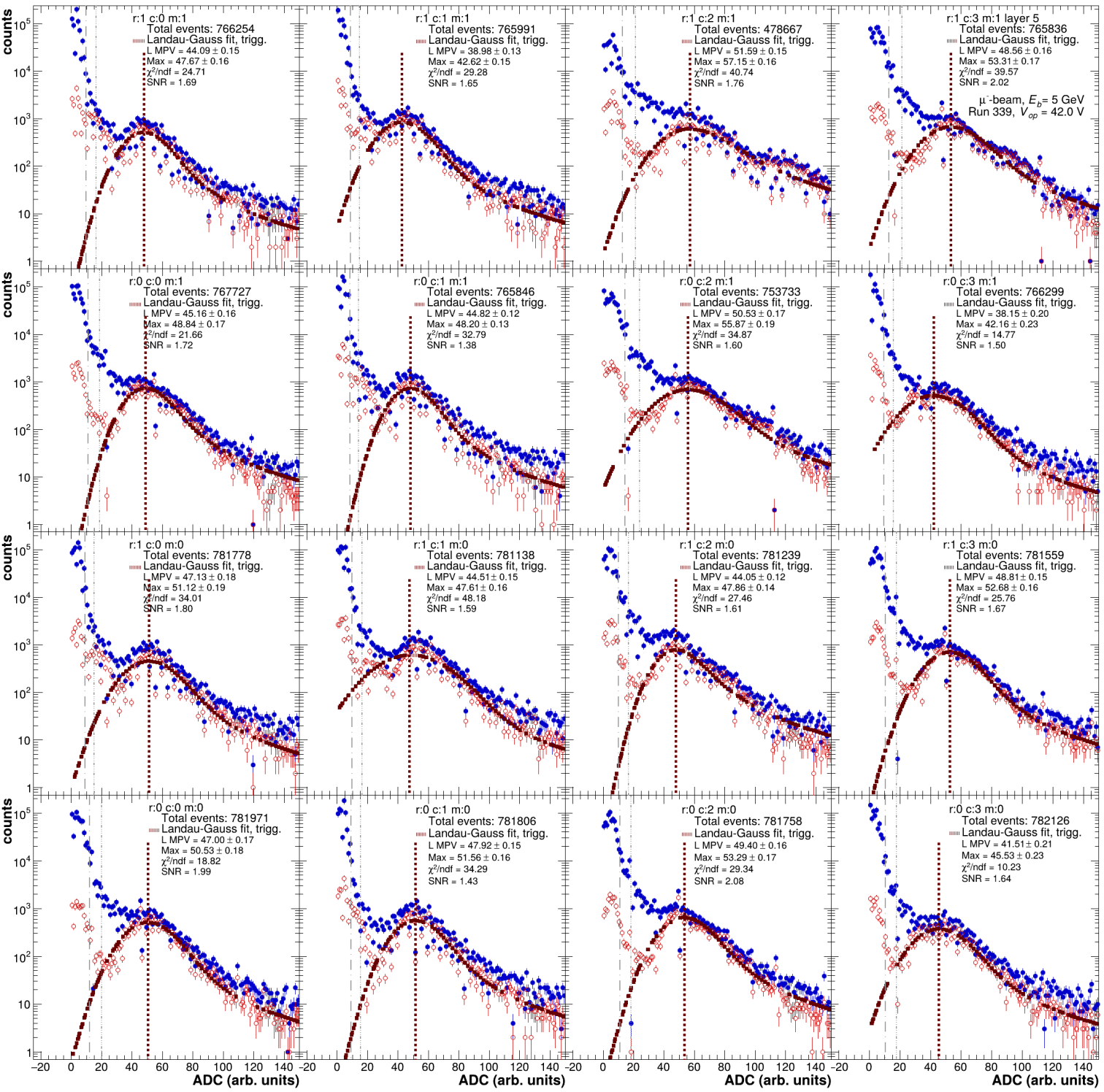

r:1 c:0 m:1
Total events: 514599
Landau-Gauss fit, trigg.
L MPV = 34.89 ± 0.10
Max = 38.54 ± 0.11
χ²/ndf = 33.83
SNR = 6.39
r:1 c:1 m:1
Total events: 477421
Landau-Gauss fit, trigg.
L MPV = 32.68 ± 0.07
Max = 35.44 ± 0.08
χ²/ndf = 65.43
SNR = 5.79
r:1 c:2 m:1
Total events: 432477
Landau-Gauss fit, trigg.
L MPV = 33.57 ± 0.08
Max = 36.63 ± 0.09
χ²/ndf = 77.80
SNR = 5.16
r:1 c:3 m:1 layer 0
Total events: 761054
Landau-Gauss fit, trigg.
L MPV = 34.94 ± 0.09
Max = 38.69 ± 0.10
χ²/ndf = 53.68
SNR = 6.29
μ-beam, E_b = 5 GeV
Run 339, V_op = 42.0 V
r:0 c:0 m:1
Total events: 513151
Landau-Gauss fit, trigg.
L MPV = 35.24 ± 0.12
Max = 39.79 ± 0.13
χ²/ndf = 36.67
SNR = 4.01
r:0 c:1 m:1
Total events: 511358
Landau-Gauss fit, trigg.
L MPV = 31.48 ± 0.10
Max = 35.77 ± 0.11
χ²/ndf = 55.54
SNR = 3.34
r:0 c:2 m:1
Total events: 475741
Landau-Gauss fit, trigg.
L MPV = 32.88 ± 0.10
Max = 37.08 ± 0.11
χ²/ndf = 59.70
SNR = 3.27
r:0 c:3 m:1
Total events: 763994
Landau-Gauss fit, trigg.
L MPV = 28.13 ± 0.09
Max = 30.90 ± 0.10
χ²/ndf = 20.05
SNR = 2.64
r:1 c:0 m:0
r:1 c:1 m:0
Total events: 778686
Landau-Gauss fit, trigg.
L MPV = 19.74 ± 0.05
Max = 20.39 ± 0.05
χ²/ndf = 68.42
SNR = 4.40
r:1 c:2 m:0
Total events: 776876
Landau-Gauss fit, trigg.
L MPV = 19.89 ± 0.05
Max = 20.69 ± 0.05
χ²/ndf = 72.60
SNR = 4.15
r:1 c:3 m:0
Total events: 780016
Landau-Gauss fit, trigg.
L MPV = 18.65 ± 0.11
Max = 20.81 ± 0.13
χ²/ndf = 69.89
SNR = 4.99
r:0 c:0 m:0
Total events: 780828
Landau-Gauss fit, trigg.
L MPV = 39.64 ± 0.00
Max = 38.64 ± 0.00
χ²/ndf = 98.98
SNR = 4.96
r:0 c:1 m:0
Total events: 780582
Landau-Gauss fit, trigg.
L MPV = 19.82 ± 0.05
Max = 20.53 ± 0.06
χ²/ndf = 63.72
SNR = 4.54
r:0 c:2 m:0
Total events: 780308
Landau-Gauss fit, trigg.
L MPV = 41.91 ± 0.01
Max = 40.91 ± 0.01
χ²/ndf = 266.24
SNR = 4.62
r:0 c:3 m:0
Total events: 781650
Landau-Gauss fit, trigg.
L MPV = 22.73 ± 0.02
Max = 22.18 ± 0.02
χ²/ndf = 38.35
SNR = 3.46
counts
ADC (arb. units)

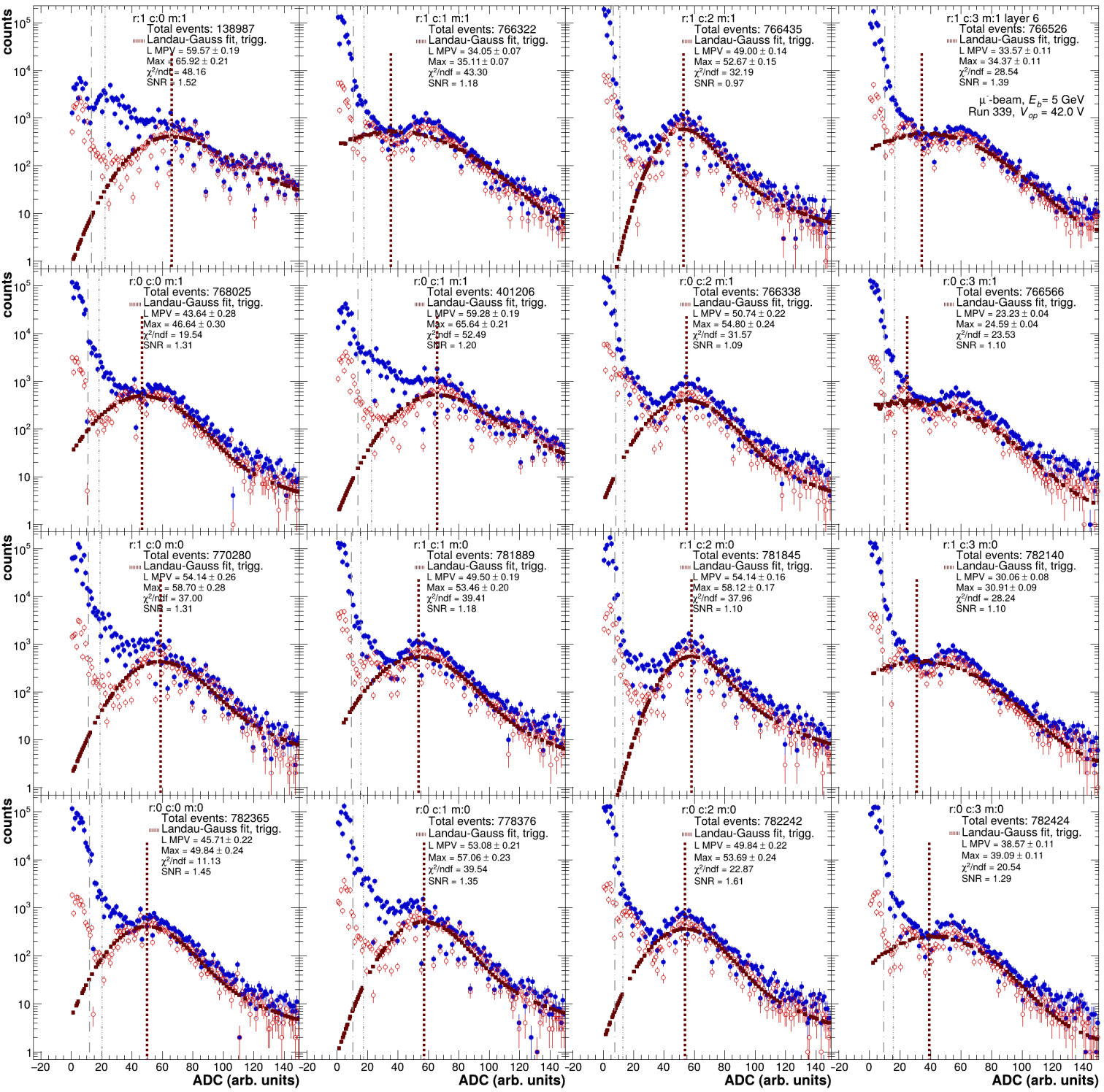

r:1 c:0 m:1
Total events: 512277
Landau-Gauss fit, trigg.
L MPV = 35.05 ± 0.10
Max = 38.71 ± 0.11
χ²/ndf = 39.50
SNR = 7.90
r:1 c:1 m:1
Total events: 745389
Landau-Gauss fit, trigg.
L MPV = 33.99 ± 0.08
Max = 37.50 ± 0.09
χ²/ndf = 44.05
SNR = 7.22
r:1 c:2 m:1
Total events: 746633
Landau-Gauss fit, trigg.
L MPV = 35.47 ± 0.08
Max = 39.40 ± 0.09
χ²/ndf = 59.61
SNR = 6.86
r:1 c:3 m:1 layer 1
Total events: 760581
Landau-Gauss fit, trigg.
L MPV = 17.45 ± 0.32
Max = 22.50 ± 0.41
χ²/ndf = 66.93
SNR = 6.03
μ-beam, E_b = 5 GeV
Run 339, V_{op} = 42.0 V
r:0 c:0 m:1
Total events: 509938
Landau-Gauss fit, trigg.
L MPV = 34.61 ± 0.11
Max = 38.73 ± 0.13
χ²/ndf = 42.12
SNR = 5.60
r:0 c:1 m:1
Total events: 745758
Landau-Gauss fit, trigg.
L MPV = 34.65 ± 0.09
Max = 38.75 ± 0.10
χ²/ndf = 51.09
SNR = 5.09
r:0 c:2 m:1
Total events: 747415
Landau-Gauss fit, trigg.
L MPV = 33.89 ± 0.09
Max = 37.85 ± 0.10
χ²/ndf = 45.41
SNR = 4.69
r:0 c:3 m:1
Total events: 762909
Landau-Gauss fit, trigg.
L MPV = 34.49 ± 0.13
Max = 38.46 ± 0.14
χ²/ndf = 18.29
SNR = 4.11
r:1 c:0 m:0
Total events: 779698
Landau-Gauss fit, trigg.
L MPV = 37.48 ± 0.01
Max = 36.51 ± 0.01
χ²/ndf = 165.21
SNR = 6.21
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 778368
Landau-Gauss fit, trigg.
L MPV = 38.63 ± 0.00
Max = 37.62 ± 0.00
χ²/ndf = 182.36
SNR = 5.85
r:1 c:3 m:0
Total events: 779630
Landau-Gauss fit, trigg.
L MPV = 16.31 ± 0.05
Max = 18.22 ± 0.06
χ²/ndf = 46.53
SNR = 5.63
r:0 c:0 m:0
r:0 c:1 m:0
Total events: 779324
Landau-Gauss fit, trigg.
L MPV = 29.69 ± 0.01
Max = 30.44 ± 0.01
χ²/ndf = 76.35
SNR = 6.48
r:0 c:2 m:0
Total events: 779915
Landau-Gauss fit, trigg.
L MPV = 40.86 ± 0.01
Max = 39.91 ± 0.01
χ²/ndf = 244.51
SNR = 6.14
r:0 c:3 m:0
counts
ADC (arb. units)

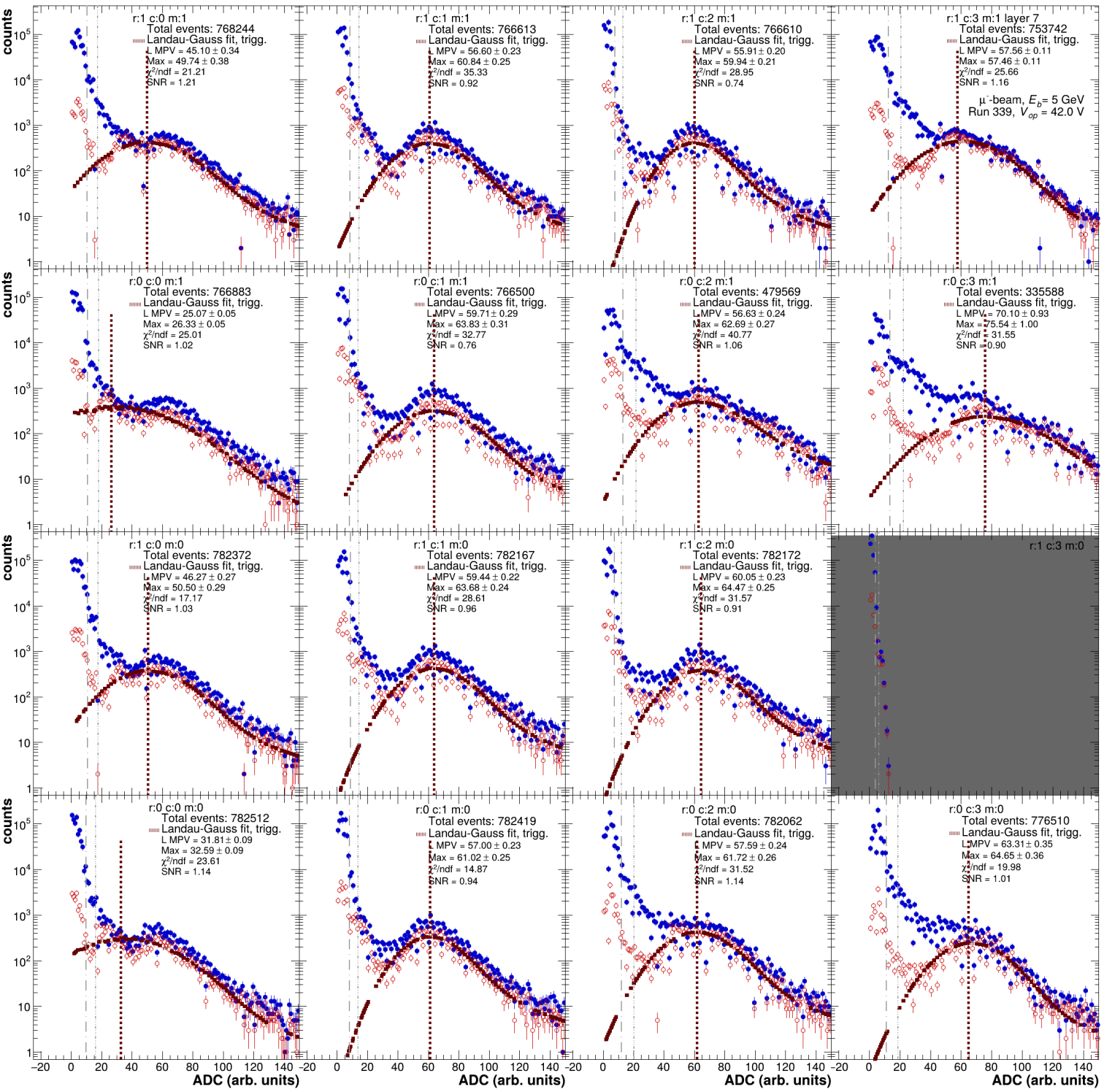

r:1 c:0 m:1
Total events: 479983
Landau-Gauss fit, trigg.
L MPV = 34.10 ± 0.10
Max = 37.90 ± 0.11
χ²/ndf = 48.25
SNR = 5.28
r:1 c:1 m:1
Total events: 760385
Landau-Gauss fit, trigg.
L MPV = 27.99 ± 0.09
Max = 31.90 ± 0.10
χ²/ndf = 38.81
SNR = 6.15
r:1 c:2 m:1
Total events: 474167
Landau-Gauss fit, trigg.
L MPV = 36.24 ± 0.10
Max = 40.69 ± 0.11
χ²/ndf = 72.71
SNR = 5.19
r:1 c:3 m:1 layer 2
Total events: 212615
Landau-Gauss fit, trigg.
L MPV = 24.97 ± 0.01
Max = 22.92 ± 0.01
χ²/ndf = 92.97
SNR = 5.20
μ-beam, E_b = 5 GeV
Run 339, V_op = 42.0 V
r:0 c:0 m:1
Total events: 515251
Landau-Gauss fit, trigg.
L MPV = 33.85 ± 0.10
Max = 37.84 ± 0.11
χ²/ndf = 41.33
SNR = 4.71
r:0 c:1 m:1
Total events: 761921
Landau-Gauss fit, trigg.
L MPV = 26.49 ± 0.11
Max = 30.71 ± 0.13
χ²/ndf = 31.67
SNR = 4.05
r:0 c:2 m:1
Total events: 762593
Landau-Gauss fit, trigg.
L MPV = 38.46 ± 0.01
Max = 37.49 ± 0.01
χ²/ndf = 146.61
SNR = 4.12
r:0 c:3 m:1
Total events: 763740
Landau-Gauss fit, trigg.
L MPV = 33.57 ± 0.12
Max = 37.47 ± 0.13
χ²/ndf = 31.52
SNR = 3.81
r:1 c:0 m:0
Total events: 777616
Landau-Gauss fit, trigg.
L MPV = 35.46 ± 0.12
Max = 39.76 ± 0.14
χ²/ndf = 38.33
SNR = 6.17
r:1 c:1 m:0
Total events: 776606
Landau-Gauss fit, trigg.
L MPV = 25.92 ± 0.07
Max = 28.80 ± 0.08
χ²/ndf = 33.60
SNR = 5.34
r:1 c:2 m:0
Total events: 775266
Landau-Gauss fit, trigg.
L MPV = 32.90 ± 0.10
Max = 36.78 ± 0.11
χ²/ndf = 40.12
SNR = 5.32
r:1 c:3 m:0
Total events: 767441
Landau-Gauss fit, trigg.
L MPV = 32.47 ± 0.12
Max = 36.19 ± 0.13
χ²/ndf = 67.47
SNR = 5.64
r:0 c:0 m:0
Total events: 779279
Landau-Gauss fit, trigg.
L MPV = 35.58 ± 0.13
Max = 39.37 ± 0.14
χ²/ndf = 24.82
SNR = 6.49
r:0 c:1 m:0
Total events: 778169
Landau-Gauss fit, trigg.
L MPV = 27.52 ± 0.09
Max = 30.97 ± 0.10
χ²/ndf = 32.20
SNR = 5.93
r:0 c:2 m:0
Total events: 778779
Landau-Gauss fit, trigg.
L MPV = 40.26 ± 0.01
Max = 39.48 ± 0.01
χ²/ndf = 178.75
SNR = 5.64
r:0 c:3 m:0
Total events: 779916
Landau-Gauss fit, trigg.
L MPV = 30.34 ± 0.10
Max = 33.29 ± 0.11
χ²/ndf = 19.21
SNR = 5.13
counts
ADC (arb. units)

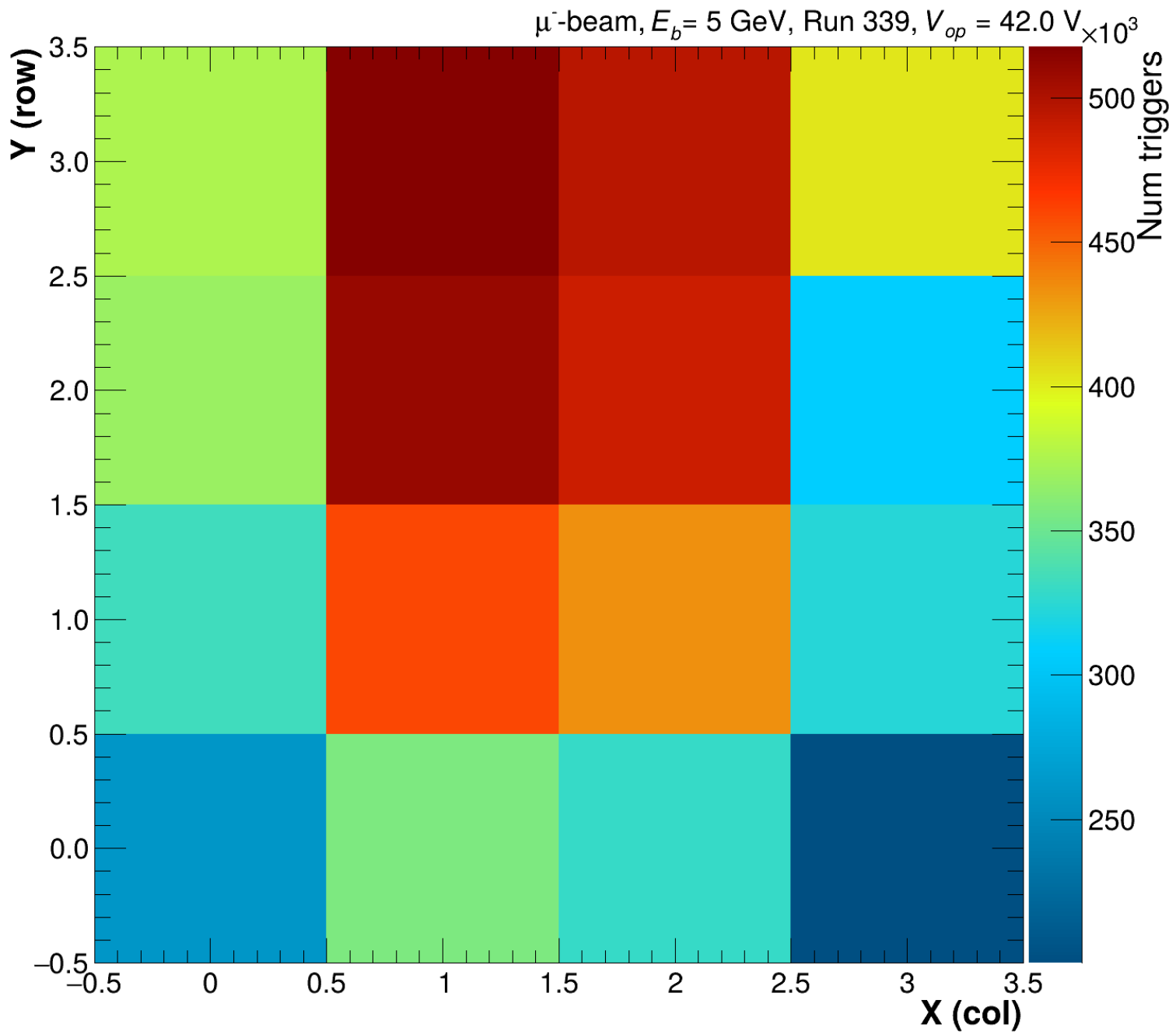

r:1 c:0 m:1
Total events: 764278
Landau-Gauss fit, trigg.
L MPV = 29.25 ± 0.13
Max = 33.12 ± 0.14
χ²/ndf = 54.87
SNR = 1.77
r:1 c:1 m:1
Total events: 283093
Landau-Gauss fit, trigg.
L MPV = 34.82 ± 0.10
Max = 39.05 ± 0.11
χ²/ndf = 83.22
SNR = 3.17
r:1 c:2 m:1
Total events: 762994
Landau-Gauss fit, trigg.
L MPV = 20.98 ± 0.23
Max = 25.48 ± 0.27
χ²/ndf = 45.56
SNR = 3.15
r:1 c:3 m:1 layer 3
Total events: 764863
Landau-Gauss fit, trigg.
L MPV = 33.05 ± 0.10
Max = 36.77 ± 0.11
χ²/ndf = 40.92
SNR = 3.50
μ⁻-beam, E_b = 5 GeV
Run 339, V_op = 42.0 V
r:0 c:0 m:1
Total events: 764142
Landau-Gauss fit, trigg.
L MPV = 27.62 ± 0.17
Max = 31.90 ± 0.20
χ²/ndf = 59.09
SNR = 2.14
r:0 c:1 m:1
Total events: 515100
Landau-Gauss fit, trigg.
L MPV = 34.16 ± 0.10
Max = 38.51 ± 0.11
χ²/ndf = 68.29
SNR = 2.94
r:0 c:2 m:1
Total events: 475133
Landau-Gauss fit, trigg.
L MPV = 31.93 ± 0.10
Max = 36.27 ± 0.12
χ²/ndf = 63.24
SNR = 2.60
r:0 c:3 m:1
Total events: 764598
Landau-Gauss fit, trigg.
L MPV = 38.07 ± 0.02
Max = 38.70 ± 0.02
χ²/ndf = 126.53
SNR = 2.48
r:1 c:0 m:0
Total events: 779425
Landau-Gauss fit, trigg.
L MPV = 32.53 ± 0.12
Max = 36.36 ± 0.14
χ²/ndf = 51.54
SNR = 1.70
r:1 c:1 m:0
Total events: 770382
Landau-Gauss fit, trigg.
L MPV = 33.17 ± 0.11
Max = 37.46 ± 0.12
χ²/ndf = 104.41
SNR = 3.06
r:1 c:2 m:0
r:1 c:3 m:0
Total events: 778476
Landau-Gauss fit, trigg.
L MPV = 32.49 ± 0.11
Max = 36.24 ± 0.12
χ²/ndf = 32.63
SNR = 3.21
r:0 c:0 m:0
Total events: 780482
Landau-Gauss fit, trigg.
L MPV = 25.69 ± 0.26
Max = 30.30 ± 0.31
χ²/ndf = 52.60
SNR = 2.30
r:0 c:1 m:0
Total events: 779404
Landau-Gauss fit, trigg.
L MPV = 34.80 ± 0.12
Max = 38.74 ± 0.13
χ²/ndf = 35.63
SNR = 3.56
r:0 c:2 m:0
Total events: 779361
Landau-Gauss fit, trigg.
L MPV = 34.88 ± 0.12
Max = 38.91 ± 0.13
χ²/ndf = 38.75
SNR = 3.25
r:0 c:3 m:0
Total events: 780417
Landau-Gauss fit, trigg.
L MPV = 27.26 ± 0.17
Max = 31.50 ± 0.20
χ²/ndf = 13.48
SNR = 2.82
counts
ADC (arb. units)

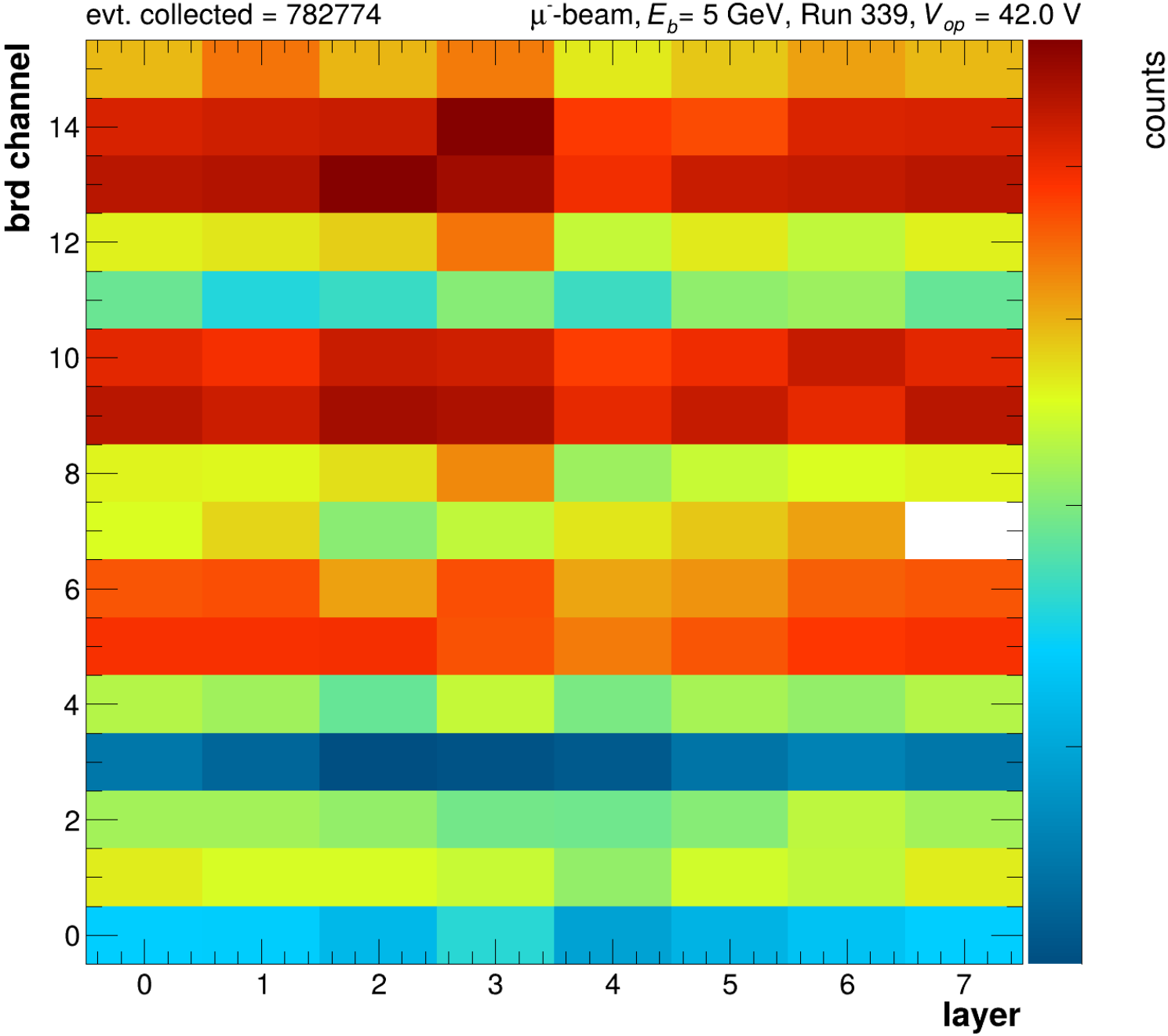

counts
ADC (arb. units)
r:1 c:0 m:1
Total events: 767003
Landau-Gauss fit, trigg.
L MPV = 40.81 ± 0.10
Max = 44.23 ± 0.11
χ²/ndf = 26.50
SNR = 2.97
r:1 c:1 m:1
Total events: 765789
Landau-Gauss fit, trigg.
L MPV = 37.37 ± 0.19
Max = 39.01 ± 0.20
χ²/ndf = 111.82
SNR = 2.61
r:1 c:2 m:1
Total events: 765037
Landau-Gauss fit, trigg.
L MPV = 36.82 ± 0.10
Max = 40.26 ± 0.10
χ²/ndf = 17.81
SNR = 2.51
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b = 5 GeV
Run 339, V_op = 42.0 V
r:0 c:0 m:1
Total events: 764985
Landau-Gauss fit, trigg.
L MPV = 48.65 ± 0.16
Max = 53.92 ± 0.18
χ²/ndf = 34.25
SNR = 2.46
r:0 c:1 m:1
Total events: 766310
Landau-Gauss fit, trigg.
L MPV = 39.99 ± 0.10
Max = 43.50 ± 0.11
χ²/ndf = 25.44
SNR = 2.14
r:0 c:2 m:1
Total events: 763962
Landau-Gauss fit, trigg.
L MPV = 45.42 ± 0.21
Max = 51.22 ± 0.23
χ²/ndf = 58.67
SNR = 2.11
r:0 c:3 m:1
Total events: 513407
Landau-Gauss fit, trigg.
L MPV = 47.93 ± 0.17
Max = 50.05 ± 0.18
χ²/ndf = 24.23
SNR = 1.95
r:1 c:0 m:0
Total events: 780806
Landau-Gauss fit, trigg.
L MPV = 42.44 ± 0.13
Max = 46.07 ± 0.14
χ²/ndf = 25.49
SNR = 2.42
r:1 c:1 m:0
Total events: 779739
Landau-Gauss fit, trigg.
L MPV = 42.26 ± 0.10
Max = 45.93 ± 0.11
χ²/ndf = 30.56
SNR = 2.46
r:1 c:2 m:0
Total events: 779993
Landau-Gauss fit, trigg.
L MPV = 39.22 ± 0.11
Max = 42.93 ± 0.12
χ²/ndf = 24.52
SNR = 2.31
r:1 c:3 m:0
Total events: 780334
Landau-Gauss fit, trigg.
L MPV = 45.78 ± 0.12
Max = 49.86 ± 0.14
χ²/ndf = 38.86
SNR = 2.55
r:0 c:0 m:0
Total events: 781493
Landau-Gauss fit, trigg.
L MPV = 46.94 ± 0.17
Max = 50.79 ± 0.18
χ²/ndf = 21.06
SNR = 2.89
r:0 c:1 m:0
Total events: 780666
Landau-Gauss fit, trigg.
L MPV = 44.83 ± 0.11
Max = 48.27 ± 0.12
χ²/ndf = 10.78
SNR = 2.43
r:0 c:2 m:0
Total events: 780836
Landau-Gauss fit, trigg.
L MPV = 45.00 ± 0.14
Max = 48.89 ± 0.15
χ²/ndf = 31.42
SNR = 2.93
r:0 c:3 m:0
Total events: 781458
Landau-Gauss fit, trigg.
L MPV = 44.59 ± 0.17
Max = 48.66 ± 0.19
χ²/ndf = 19.68
SNR = 2.43

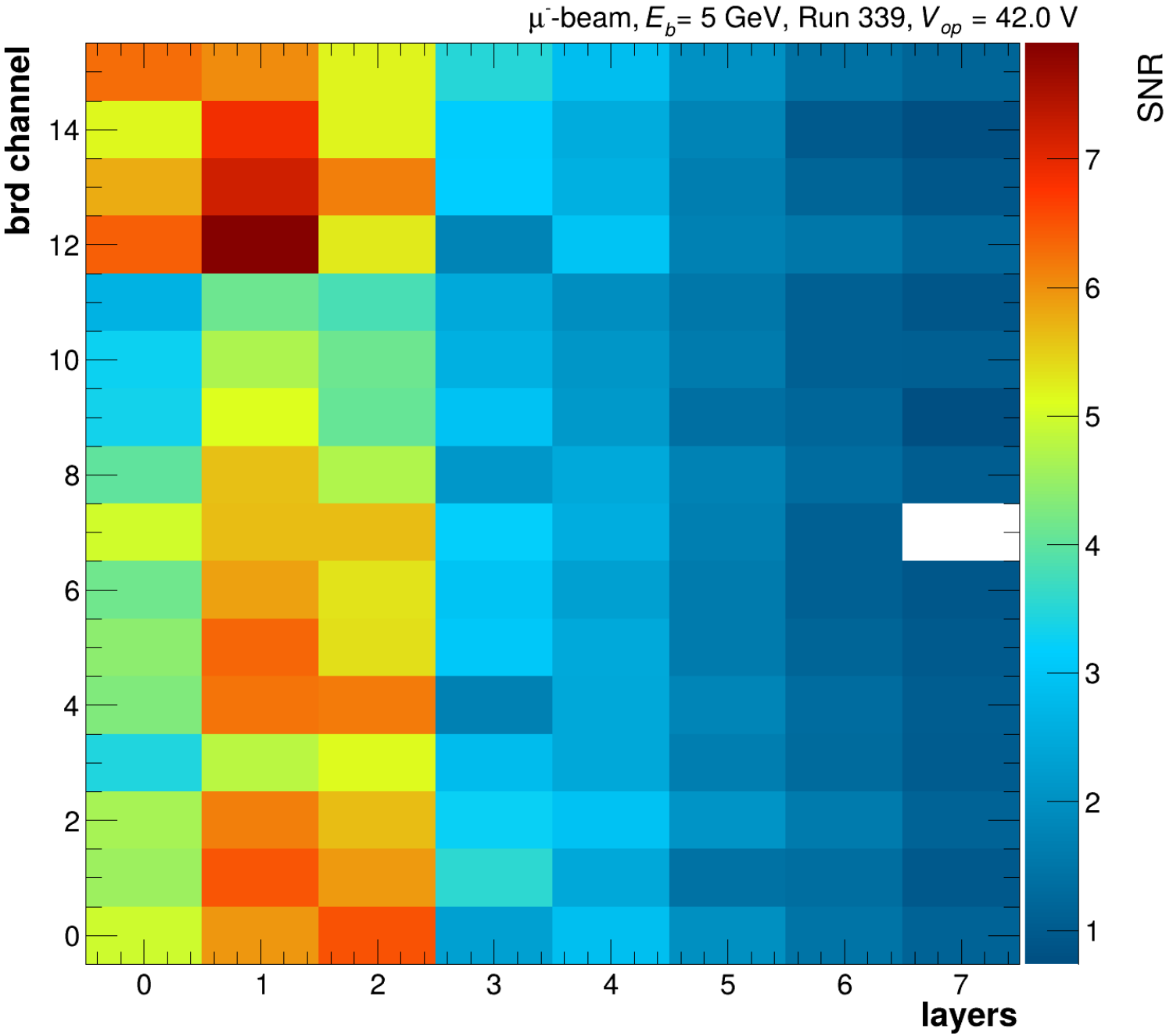

r:1 c:0 m:1
Total events: 766254
Landau-Gauss fit, trigg.
L MPV = 44.09 ± 0.15
Max = 47.67 ± 0.16
χ²/ndf = 24.71
SNR = 1.69
r:1 c:1 m:1
Total events: 765991
Landau-Gauss fit, trigg.
L MPV = 38.98 ± 0.13
Max = 42.62 ± 0.15
χ²/ndf = 29.28
SNR = 1.65
r:1 c:2 m:1
Total events: 478667
Landau-Gauss fit, trigg.
L MPV = 51.59 ± 0.15
Max = 57.15 ± 0.16
χ²/ndf = 40.74
SNR = 1.76
r:1 c:3 m:1 layer 5
Total events: 765836
Landau-Gauss fit, trigg.
L MPV = 48.56 ± 0.16
Max = 53.31 ± 0.17
χ²/ndf = 39.57
SNR = 2.02
μ-beam, E_b = 5 GeV
Run 339, V_{op} = 42.0 V
r:0 c:0 m:1
Total events: 767727
Landau-Gauss fit, trigg.
L MPV = 45.16 ± 0.16
Max = 48.84 ± 0.17
χ²/ndf = 21.66
SNR = 1.72
r:0 c:1 m:1
Total events: 765846
Landau-Gauss fit, trigg.
L MPV = 44.82 ± 0.12
Max = 48.20 ± 0.13
χ²/ndf = 32.79
SNR = 1.38
r:0 c:2 m:1
Total events: 753733
Landau-Gauss fit, trigg.
L MPV = 50.53 ± 0.17
Max = 55.87 ± 0.19
χ²/ndf = 34.87
SNR = 1.60
r:0 c:3 m:1
Total events: 766299
Landau-Gauss fit, trigg.
L MPV = 38.15 ± 0.20
Max = 42.16 ± 0.23
χ²/ndf = 14.77
SNR = 1.50
r:1 c:0 m:0
Total events: 781778
Landau-Gauss fit, trigg.
L MPV = 47.13 ± 0.18
Max = 51.12 ± 0.19
χ²/ndf = 34.01
SNR = 1.80
r:1 c:1 m:0
Total events: 781138
Landau-Gauss fit, trigg.
L MPV = 44.51 ± 0.15
Max = 47.61 ± 0.16
χ²/ndf = 48.18
SNR = 1.59
r:1 c:2 m:0
Total events: 781239
Landau-Gauss fit, trigg.
L MPV = 44.05 ± 0.12
Max = 47.86 ± 0.14
χ²/ndf = 27.46
SNR = 1.61
r:1 c:3 m:0
Total events: 781559
Landau-Gauss fit, trigg.
L MPV = 48.81 ± 0.15
Max = 52.68 ± 0.16
χ²/ndf = 25.76
SNR = 1.67
r:0 c:0 m:0
Total events: 781971
Landau-Gauss fit, trigg.
L MPV = 47.00 ± 0.17
Max = 50.53 ± 0.18
χ²/ndf = 18.82
SNR = 1.99
r:0 c:1 m:0
Total events: 781806
Landau-Gauss fit, trigg.
L MPV = 47.92 ± 0.15
Max = 51.56 ± 0.16
χ²/ndf = 34.29
SNR = 1.43
r:0 c:2 m:0
Total events: 781758
Landau-Gauss fit, trigg.
L MPV = 49.40 ± 0.16
Max = 53.29 ± 0.17
χ²/ndf = 29.34
SNR = 2.08
r:0 c:3 m:0
Total events: 782126
Landau-Gauss fit, trigg.
L MPV = 41.51 ± 0.21
Max = 45.53 ± 0.23
χ²/ndf = 10.23
SNR = 1.64
counts
ADC (arb. units)

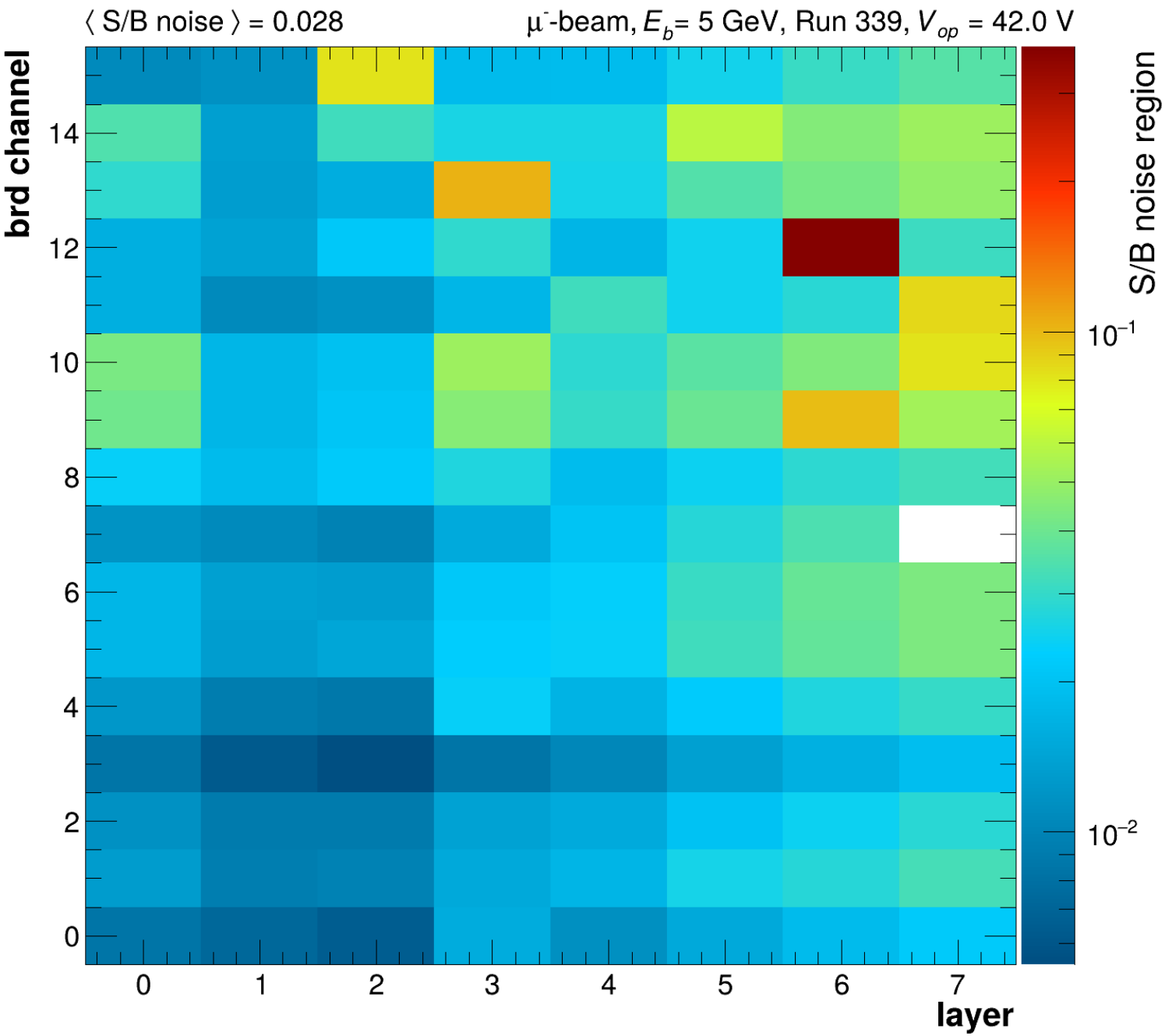

r:1 c:0 m:1
Total events: 138987
Landau-Gauss fit, trigg.
L MPV = 59.57 ± 0.19
Max = 65.92 ± 0.21
χ²/ndf = 48.16
SNR = 1.52
r:1 c:1 m:1
Total events: 766322
Landau-Gauss fit, trigg.
L MPV = 34.05 ± 0.07
Max = 35.11 ± 0.07
χ²/ndf = 43.30
SNR = 1.18
r:1 c:2 m:1
Total events: 766435
Landau-Gauss fit, trigg.
L MPV = 49.00 ± 0.14
Max = 52.67 ± 0.15
χ²/ndf = 32.19
SNR = 0.97
r:1 c:3 m:1 layer 6
Total events: 766526
Landau-Gauss fit, trigg.
L MPV = 33.57 ± 0.11
Max = 34.37 ± 0.11
χ²/ndf = 28.54
SNR = 1.39
μ⁻-beam, E_b = 5 GeV
Run 339, V_op = 42.0 V
r:0 c:0 m:1
Total events: 768025
Landau-Gauss fit, trigg.
L MPV = 43.64 ± 0.28
Max = 46.64 ± 0.30
χ²/ndf = 19.54
SNR = 1.31
r:0 c:1 m:1
Total events: 401206
Landau-Gauss fit, trigg.
L MPV = 59.28 ± 0.19
Max = 65.64 ± 0.21
χ²/ndf = 52.49
SNR = 1.20
r:0 c:2 m:1
Total events: 766338
Landau-Gauss fit, trigg.
L MPV = 50.74 ± 0.22
Max = 54.80 ± 0.24
χ²/ndf = 31.57
SNR = 1.09
r:0 c:3 m:1
Total events: 766566
Landau-Gauss fit, trigg.
L MPV = 23.23 ± 0.04
Max = 24.59 ± 0.04
χ²/ndf = 23.53
SNR = 1.10
r:1 c:0 m:0
Total events: 770280
Landau-Gauss fit, trigg.
L MPV = 54.14 ± 0.26
Max = 58.70 ± 0.28
χ²/ndf = 37.00
SNR = 1.31
r:1 c:1 m:0
Total events: 781889
Landau-Gauss fit, trigg.
L MPV = 49.50 ± 0.19
Max = 53.46 ± 0.20
χ²/ndf = 39.41
SNR = 1.18
r:1 c:2 m:0
Total events: 781845
Landau-Gauss fit, trigg.
L MPV = 54.14 ± 0.16
Max = 58.12 ± 0.17
χ²/ndf = 37.96
SNR = 1.10
r:1 c:3 m:0
Total events: 782140
Landau-Gauss fit, trigg.
L MPV = 30.06 ± 0.08
Max = 30.91 ± 0.09
χ²/ndf = 28.24
SNR = 1.10
r:0 c:0 m:0
Total events: 782365
Landau-Gauss fit, trigg.
L MPV = 45.71 ± 0.22
Max = 49.84 ± 0.24
χ²/ndf = 11.13
SNR = 1.45
r:0 c:1 m:0
Total events: 778376
Landau-Gauss fit, trigg.
L MPV = 53.08 ± 0.21
Max = 57.06 ± 0.23
χ²/ndf = 39.54
SNR = 1.35
r:0 c:2 m:0
Total events: 782242
Landau-Gauss fit, trigg.
L MPV = 49.84 ± 0.22
Max = 53.69 ± 0.24
χ²/ndf = 22.87
SNR = 1.61
r:0 c:3 m:0
Total events: 782424
Landau-Gauss fit, trigg.
L MPV = 38.57 ± 0.11
Max = 39.09 ± 0.11
χ²/ndf = 20.54
SNR = 1.29
counts
ADC (arb. units)

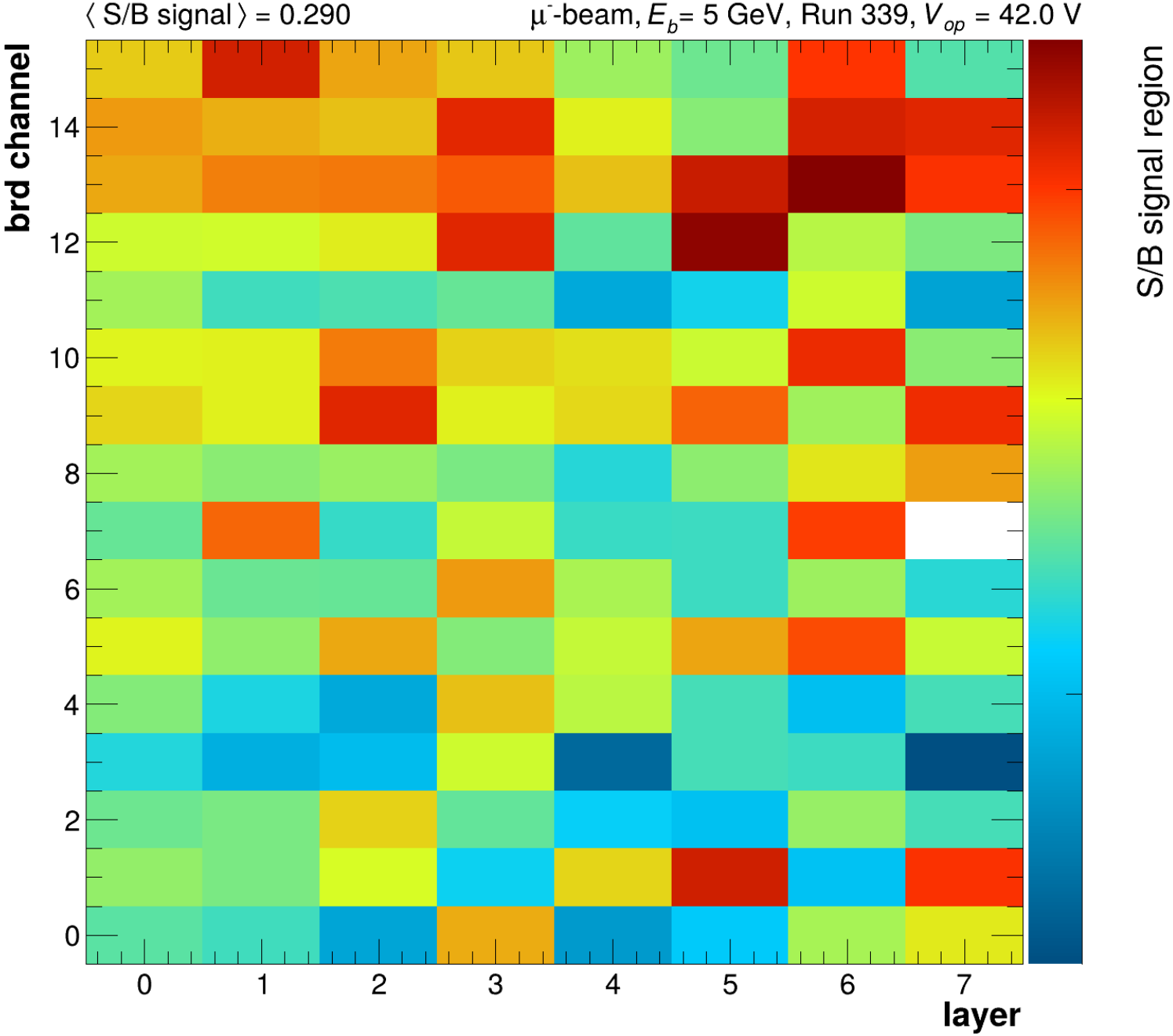

r:1 c:0 m:1
Total events: 768244
Landau-Gauss fit, trigg.
L MPV = 45.10 ± 0.34
Max = 49.74 ± 0.38
χ²/ndf = 21.21
SNR = 1.21
r:1 c:1 m:1
Total events: 766613
Landau-Gauss fit, trigg.
L MPV = 56.60 ± 0.23
Max = 60.84 ± 0.25
χ²/ndf = 35.33
SNR = 0.92
r:1 c:2 m:1
Total events: 766610
Landau-Gauss fit, trigg.
L MPV = 55.91 ± 0.20
Max = 59.94 ± 0.21
χ²/ndf = 28.95
SNR = 0.74
r:1 c:3 m:1 layer 7
Total events: 753742
Landau-Gauss fit, trigg.
L MPV = 57.56 ± 0.11
Max = 57.46 ± 0.11
χ²/ndf = 25.66
SNR = 1.16
μ-beam, E_b = 5 GeV
Run 339, V_{op} = 42.0 V
r:0 c:0 m:1
Total events: 766883
Landau-Gauss fit, trigg.
L MPV = 25.07 ± 0.05
Max = 26.33 ± 0.05
χ²/ndf = 25.01
SNR = 1.02
r:0 c:1 m:1
Total events: 766500
Landau-Gauss fit, trigg.
L MPV = 59.71 ± 0.29
Max = 63.83 ± 0.31
χ²/ndf = 32.77
SNR = 0.76
r:0 c:2 m:1
Total events: 479569
Landau-Gauss fit, trigg.
L MPV = 56.63 ± 0.24
Max = 62.69 ± 0.27
χ²/ndf = 40.77
SNR = 1.06
r:0 c:3 m:1
Total events: 335588
Landau-Gauss fit, trigg.
L MPV = 70.10 ± 0.93
Max = 75.54 ± 1.00
χ²/ndf = 31.55
SNR = 0.90
r:1 c:0 m:0
Total events: 782372
Landau-Gauss fit, trigg.
L MPV = 46.27 ± 0.27
Max = 50.50 ± 0.29
χ²/ndf = 17.17
SNR = 1.03
r:1 c:1 m:0
Total events: 782167
Landau-Gauss fit, trigg.
L MPV = 59.44 ± 0.22
Max = 63.68 ± 0.24
χ²/ndf = 28.61
SNR = 0.96
r:1 c:2 m:0
Total events: 782172
Landau-Gauss fit, trigg.
L MPV = 60.05 ± 0.23
Max = 64.47 ± 0.25
χ²/ndf = 31.57
SNR = 0.91
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 782512
Landau-Gauss fit, trigg.
L MPV = 31.81 ± 0.09
Max = 32.59 ± 0.09
χ²/ndf = 23.61
SNR = 1.14
r:0 c:1 m:0
Total events: 782419
Landau-Gauss fit, trigg.
L MPV = 57.00 ± 0.23
Max = 61.02 ± 0.25
χ²/ndf = 14.87
SNR = 0.94
r:0 c:2 m:0
Total events: 782062
Landau-Gauss fit, trigg.
L MPV = 57.59 ± 0.24
Max = 61.72 ± 0.26
χ²/ndf = 31.52
SNR = 1.14
r:0 c:3 m:0
Total events: 776510
Landau-Gauss fit, trigg.
L MPV = 63.31 ± 0.35
Max = 64.65 ± 0.36
χ²/ndf = 19.98
SNR = 1.01
counts
ADC (arb. units)

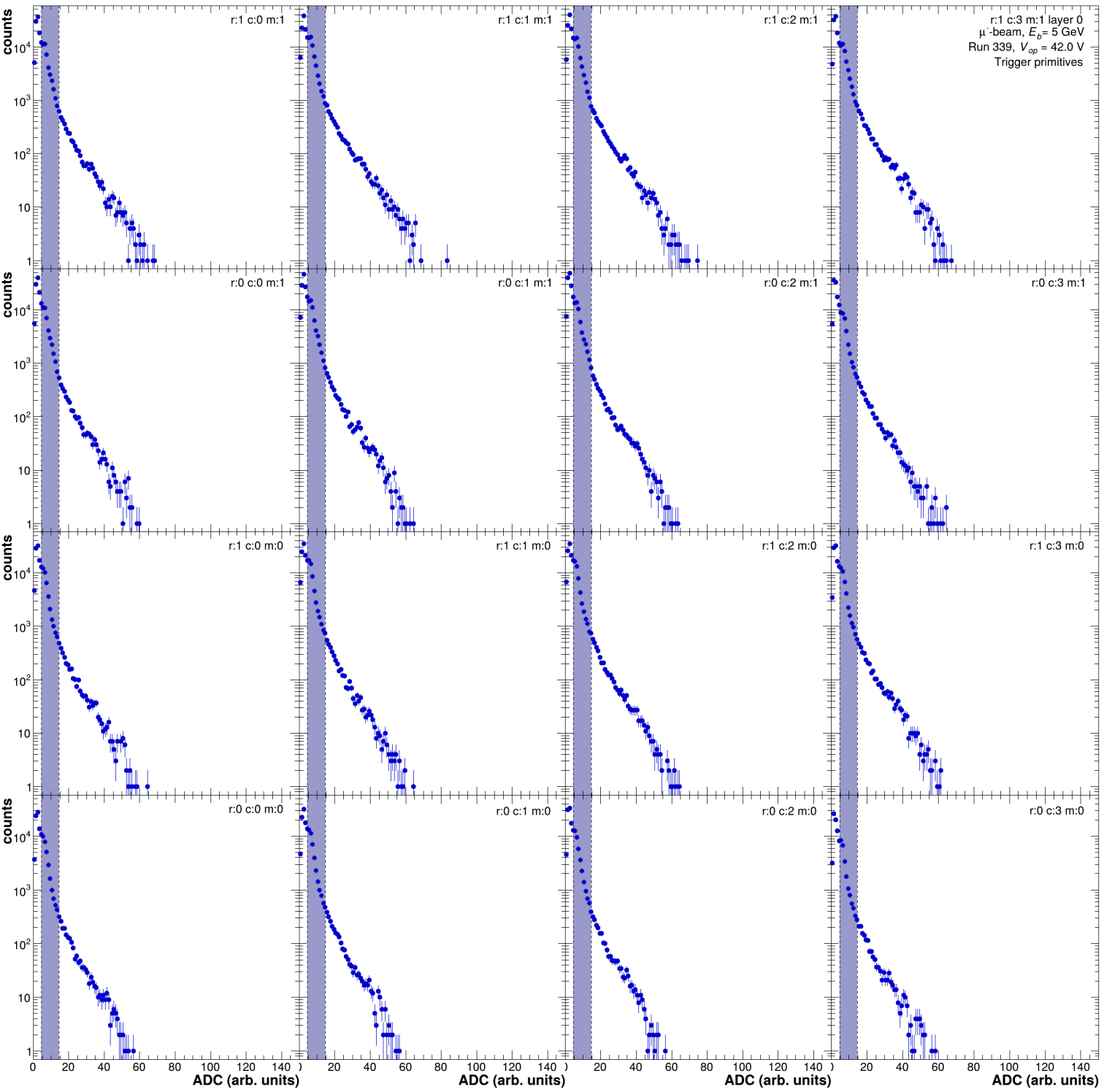

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
500
450
400
350
300
250

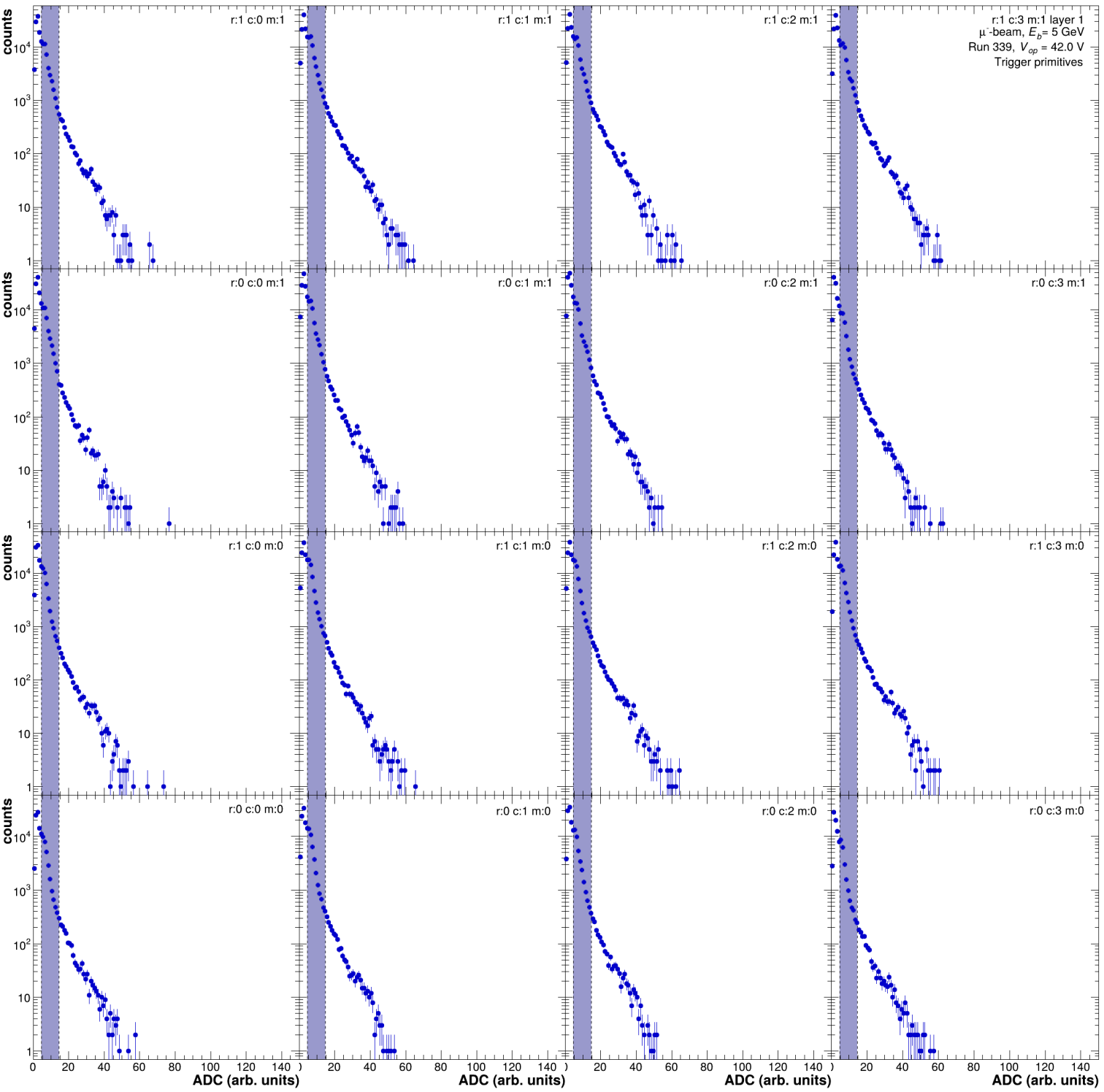

evt. collected = 782774
μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

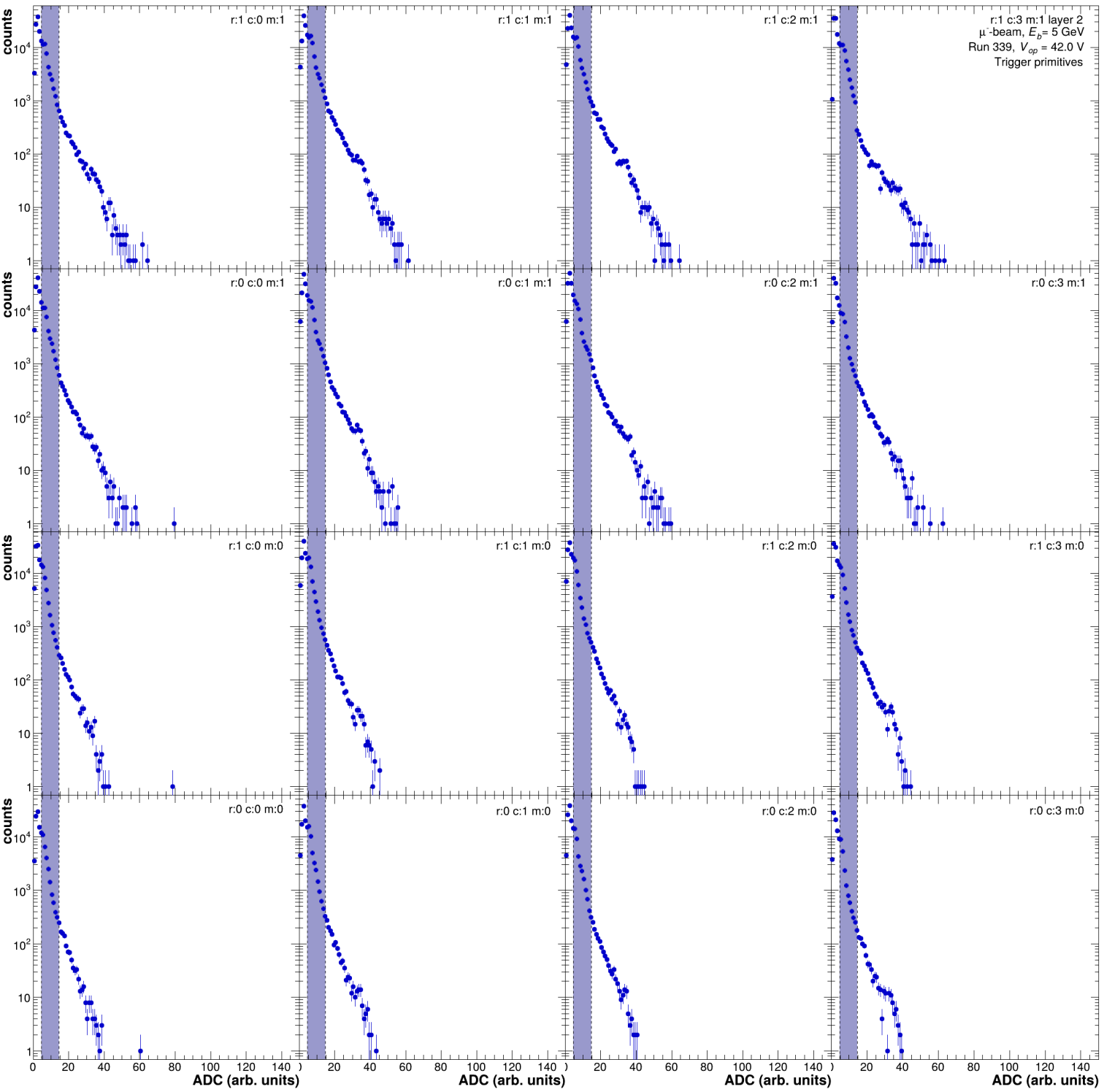

μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layers
SNR
1
2
3
4
5
6
7

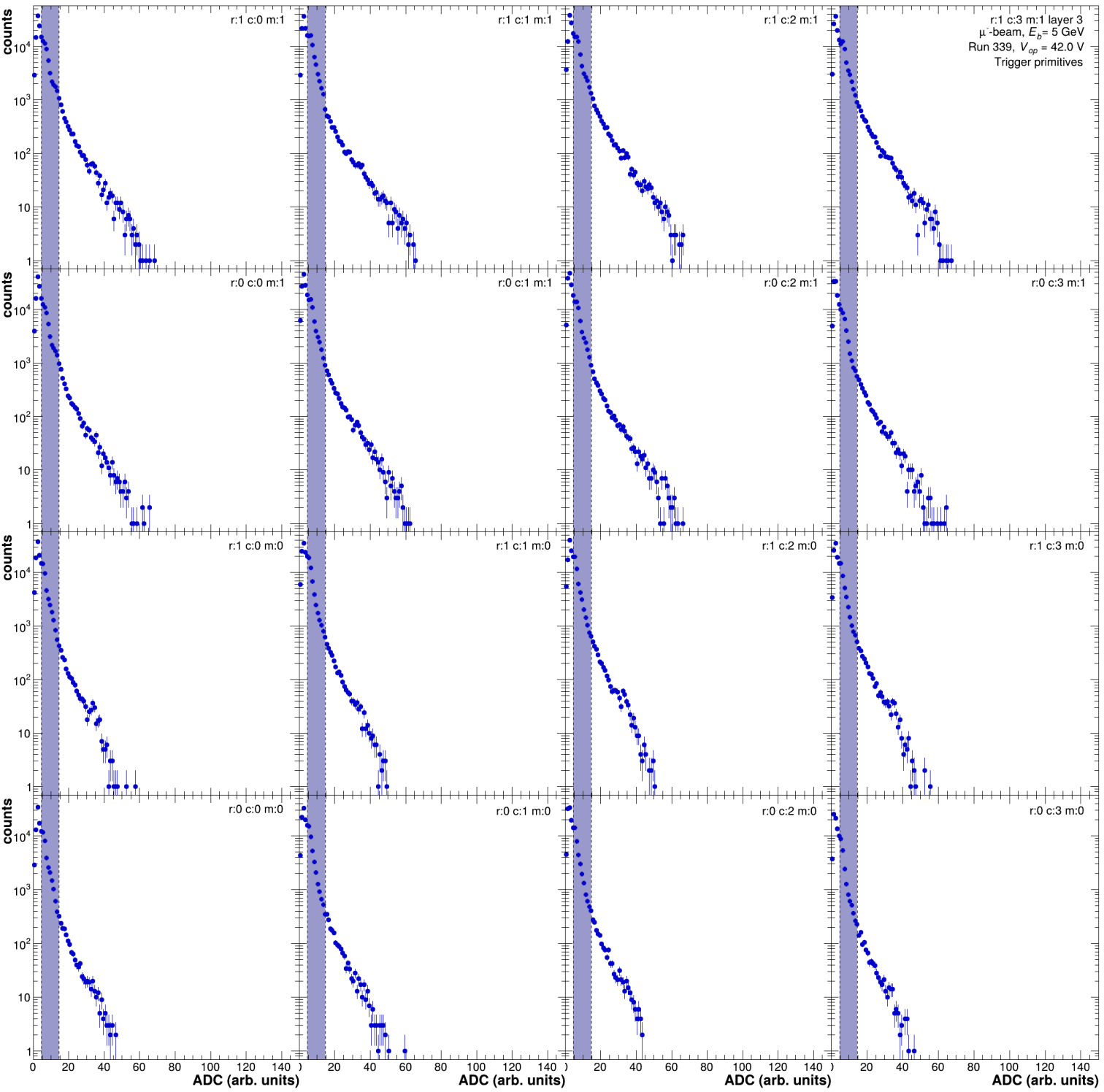

⟨ S/B noise ⟩ = 0.028
μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

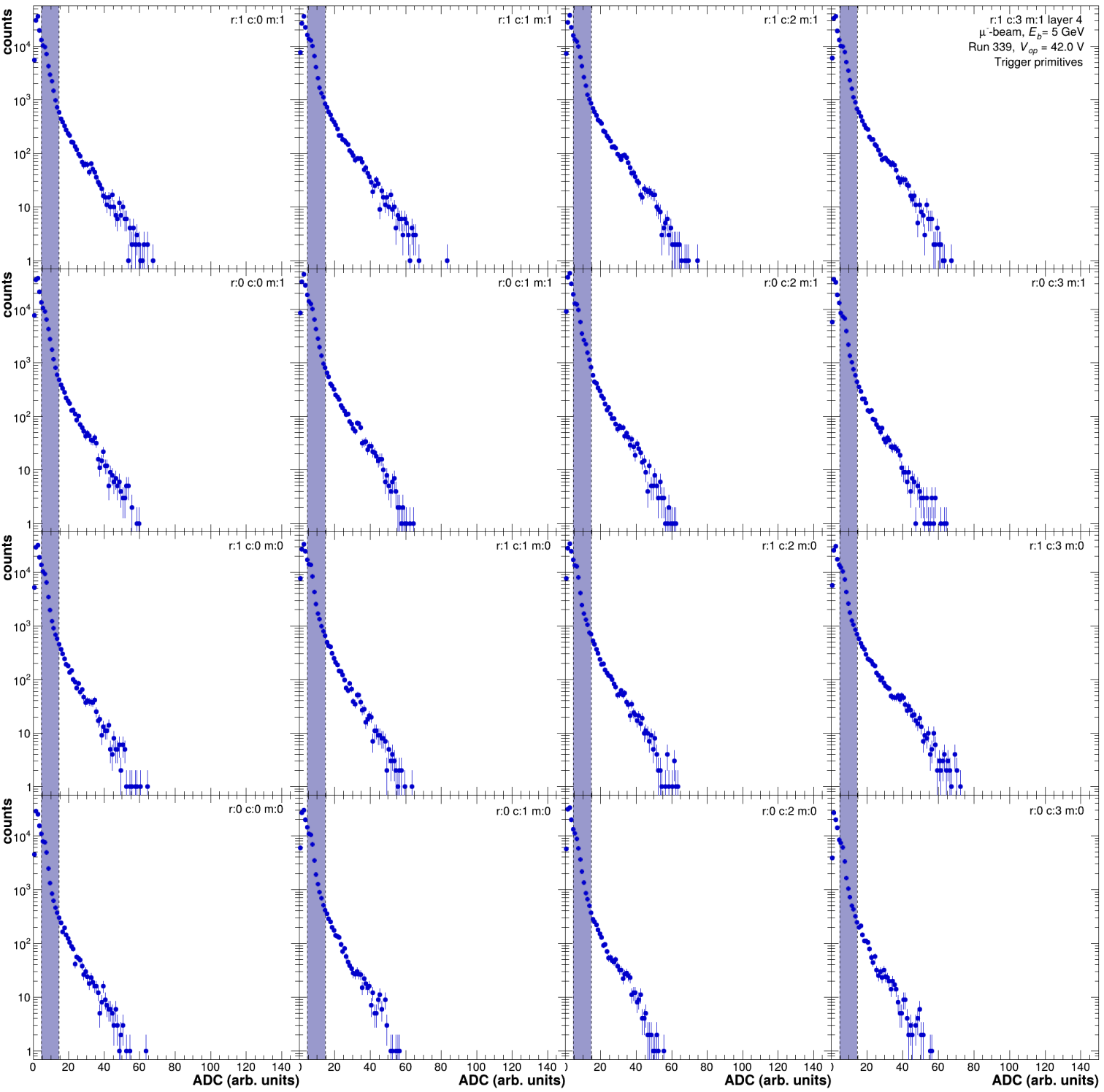

⟨ S/B signal ⟩ = 0.290
μ⁻-beam, E_b= 5 GeV, Run 339, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region

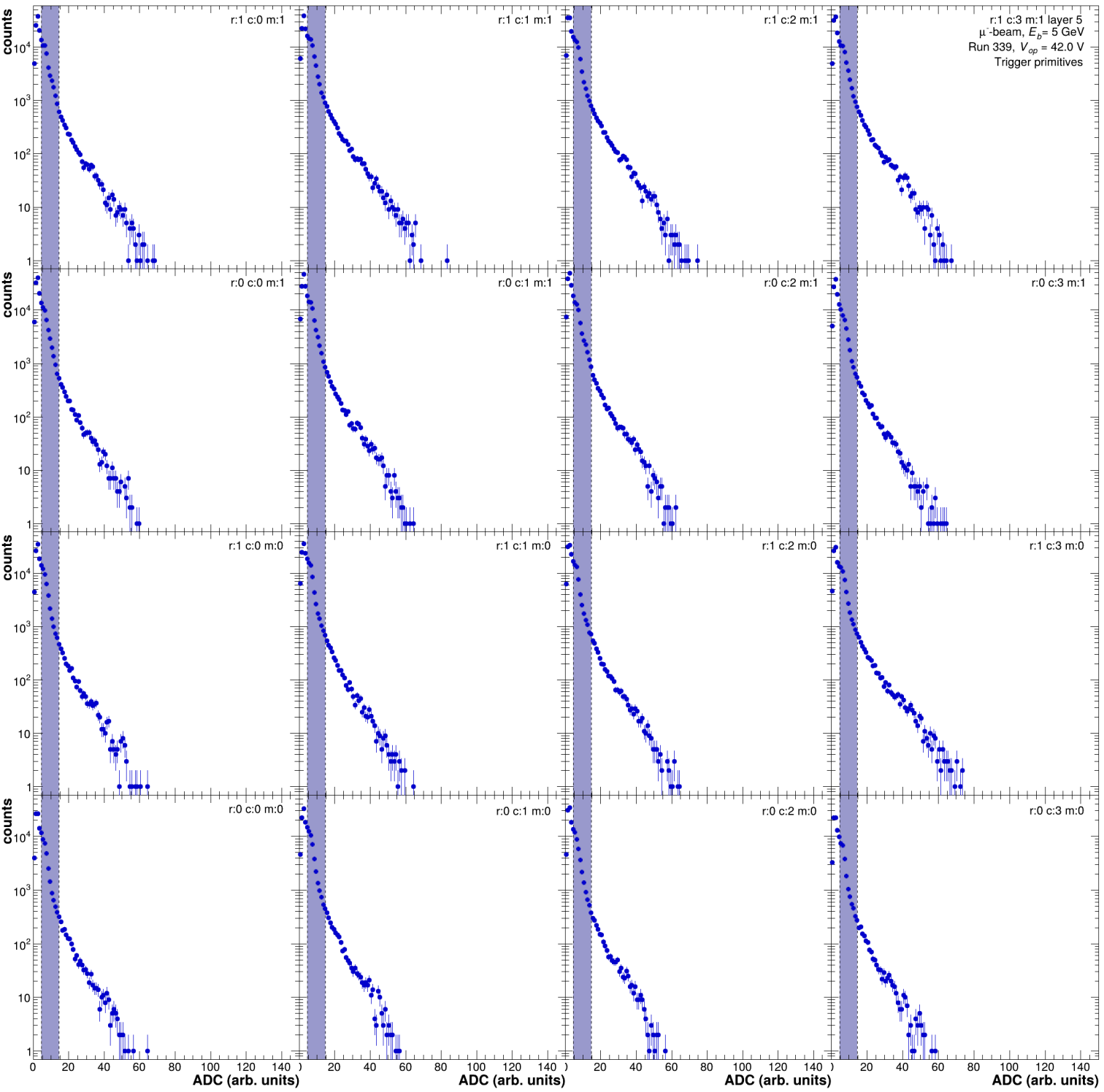

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

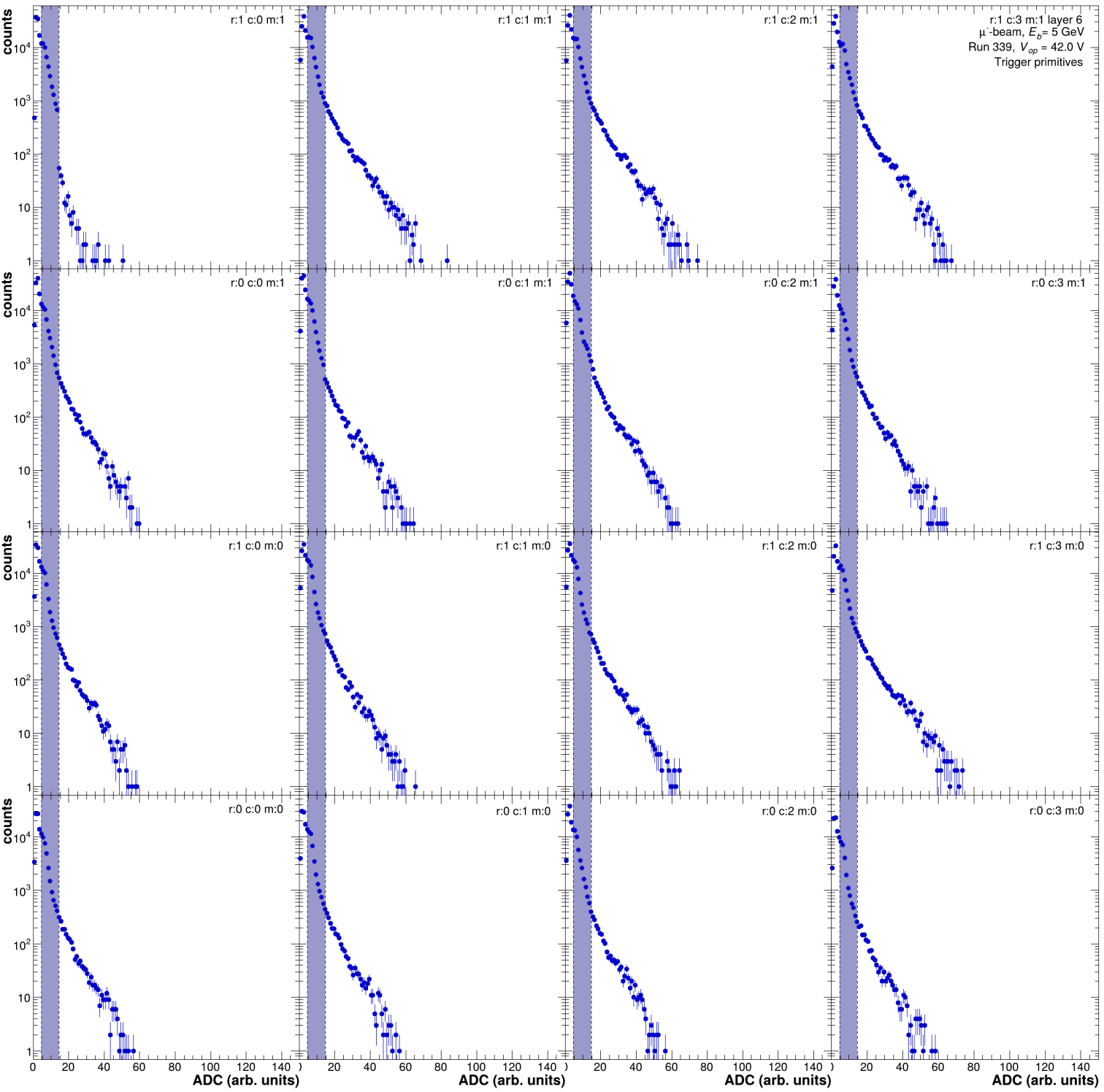

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

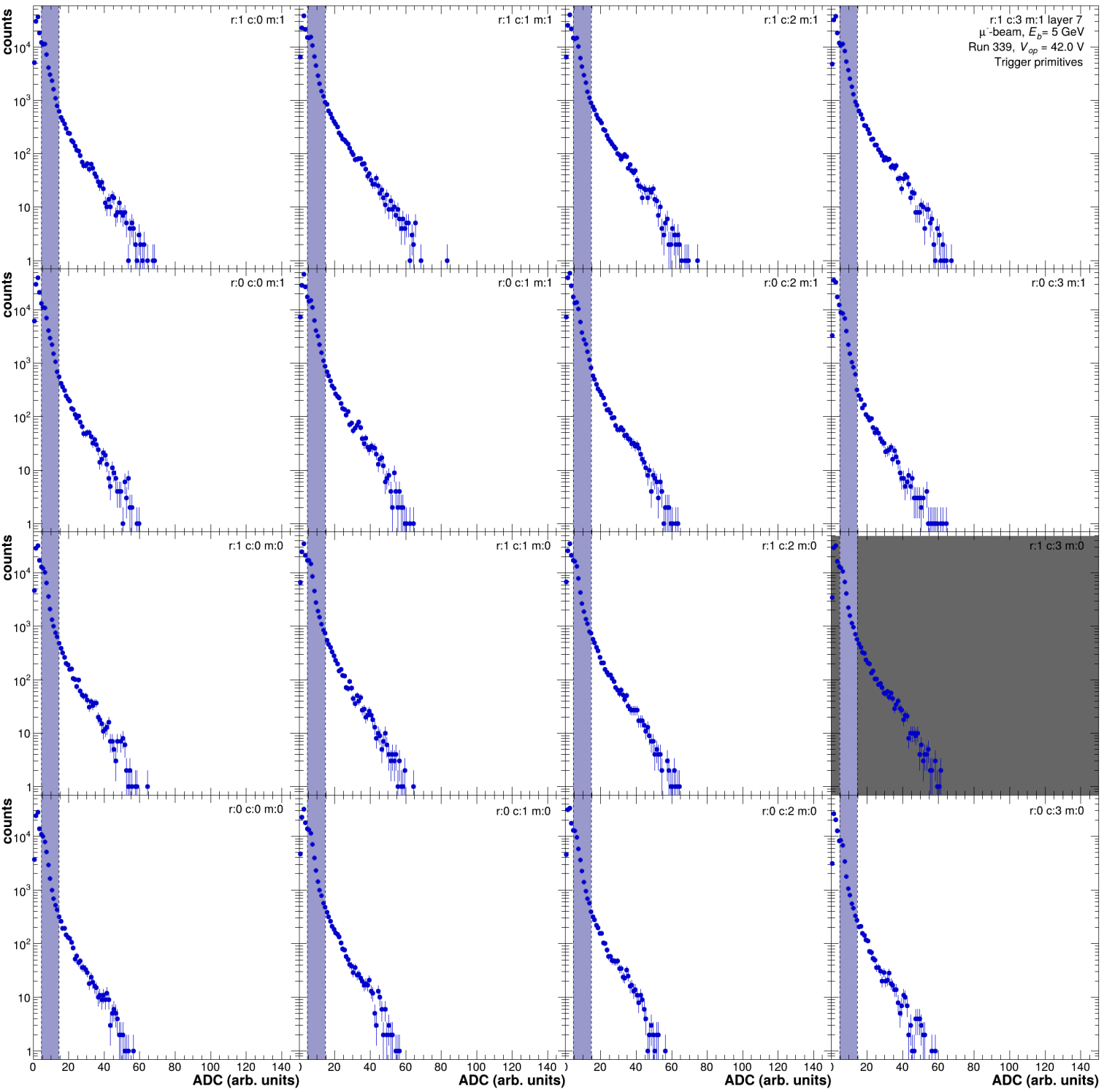

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 339, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)